SAINT MICHAEL COLLEGE OF CARAGA

RESEARCH SUPPORT SERVICES AND MONITORING SYSTEM
FOR SAINT MICHAEL COLLEGE OF CARAGA

A Thesis Presented To
The faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BATCHELOR OF SCIENCE AND INFORMATION TECHNOLOGY

By

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MARCH 2026

ACKNOWLEDGEMENT

The researchers would like to express their heartfelt gratitude to all the individuals who contributed to the success of this study, entitled “Research Support Services and Monitoring System for Saint Michael College of Caraga.”

First and foremost, our sincere appreciation goes to Mrs. Lealil I. Palacio, MSIT (CAR), our research adviser, for her valuable guidance, encouragement, and continuous support throughout the development of this study. Her expertise and patience have been a great help to us in completing this research.

The researcher would also like to extend our deepest thanks to our system beneficiary, Mr. Kenneth Ian Barrera, for his cooperation, valuable insights, and feedback that greatly contributed to the improvement of our system.

Special thanks are also given to the Learning Resource Center of Saint Michael College of Caraga for conducting research seminars that provided us with essential knowledge and guidance in completing this study.

Lastly, we are truly grateful to our families, friends, and classmates for their unending support, understanding, and motivation. Above all, we give thanks to Almighty God for the wisdom, strength, and perseverance that made this research possible.

With sincere gratitude,

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CHAPTER 1

INTRODUCTION

1.1 Project Context

Saint Michael College of Caraga is constantly working to improve the research level by improving services for faculty and student researchers. Some of the things that research aid does now, like checking for grammar mistakes, doing statistical analysis, finding plagiarism, and doing an ethical review, are still done by hand and one by one by each staff member. A system that is so disjointed causes delays, bad communication, and students, advisors, and study staff who do not work together.

With the rise of modern technology and the increasing need for more efficient research processes, many institutions already use digital platforms to support management and academic services. Studies have shown that using research support systems helps improve coordination, documentation, and service delivery in educational institutions [1]. Despite this, SMCC continues to face challenges in maintaining an orderly, centralized process, as most research transactions rely on manual methods.

Researchers developed the Research Support Services and Monitoring System (RSSMS) to solve these issues. It is a centralized web-based tool that automates and streamlines research support tasks. Students can send papers and requests directly to specific staff members, like advisers, human grammarians, statisticians, librarians, and ethics officers. The Administrator (Research Director) oversees all transactions and is present at all stages, from submission to review permission and certification [2].

RSSMS empowers everyone involved in the study to collaborate more quickly and clearly. By using this system, researchers can eliminate the need for paper routings, and they will record and track all submissions, notes, and approvals in real time [3]. We make people more accountable, reduce process time, and improve the general management of research services at the institution by using integrated features like monitoring, communication[4], and notification systems [4, [5].

Further, implementing RSSMS will make research processes more efficient and encourage students and faculty to be more involved. Streamlining the work and putting in place real-time tracking will cut down on mistakes made by people, speed up feedback to advisors, and make it easier for researchers and support staff to work together. These changes will help SMCC keep up its standards for openness, timeliness, and digital transformation in research management. The project's goal is to create a digital space that works well and is trustworthy, which will support SMCC's dedication to academic success and research integrity [6].

1.2 Literature Review

The Research Support Services and Monitoring System (RSSMS) draws on theoretical and practical foundations from studies and literature presented in this chapter. The research examines five main topics: digital research support, evolution and web-based monitoring systems, and institutional requirements for centralized solutions, real-time transparency, and workflow automation.

The Evolution of Research Support: From Traditional Services to Integrated Digital Ecosystems

Academic institutions now use digital ecosystems that integrate physical models to support students and researchers innovatively. The data show that the pandemic taught universities how to keep their study going by sending e-books, e-journals, and digital references

[7]. The study showed that the move to digital was successful. However, it also showed two main problems: insufficient digital infrastructure and students who did not fully use digital tools. To meet the identified needs, the suggested solution includes features for automated monitoring, analytics, and user support. The Research Support Services and Monitoring System for Saint Michael College of Caraga (RSSMS)

needs high-tech tools.

As a necessary step forward, academic institutions have adopted ICT-based research support [8]tools [8]. Technology such as plagiarism detectors, citation and reference managers, and Research Data Management (RDM) programs assists researchers in conducting academic research. Technology integration has changed academic support from reactive service delivery to intelligent, automatic, and proactive systems. With RSSMS's development, it is necessary to use automated tracking systems and AI-based tools to improve research operations and keep up with compliance standards and service coordination a cross department.

Two important factors drive research support for successful change: institutions that embrace readiness and individuals with the skills to use technology effectively. The study in [9] showed that institutions must create digital literacy policies and build strong partnerships with technology companies to achieve long-lasting research management innovation. The framework lets RSSMS become a model for institutional readiness because it helps researchers learn how to use technology well, has a smooth research workflow, and has a method for managing feedback and documents.

The hybrid research support model was developed by [10]. Integrates virtual and physical contact to create an environment that boosts research efficiency while making resources more accessible. The model demonstrates how researchers can benefit from automated tools while receiving individualized guidance through human interaction. The hybrid approach of RSSMS enables researchers to benefit from digital transaction management while preserving essential opportunities for academic collaboration between students and faculty members.

According to studies, the research ecosystem has transitioned from separate systems to unified technology-based systems [11]. The Research Support Services and Monitoring System for Saint Michael College of Caraga (RSSMS) will achieve digital research management unification and accessibility improvements, enhanced monitoring, and adaptive environment development through these established principles.

Web-Based Systems for Academic Monitoring and Management

Academic institutions now use web-based systems to monitor and control their operations through digital platforms. The Online Faculty Monitoring and Evaluation System developed in [12] by a nearby Philippine college showed how digital platforms improve operational efficiency, data accuracy, and transparency [13]. Despite this, SMCC faces challenges in maintaining an orderly, centralized process because most researchers continue to conduct transactions manually.

To address these problems, researchers developed the Research Support Service, which included a Web-Based Attendance Monitoring System that used open-source web technologies to improve accessibility and reduce manual recording errors. Although the primary function was attendance checking, its technical architecture, including centralized databases and real-time tracking, offers valuable insights for designing the RSSMS. Researchers can modify these web-based systems to support ongoing monitoring of research activities, automate submissions, and produce reports for institutional use. This strategy ensures a more organized and trustworthy digital monitoring procedure that aligns with academic management requirements.

From a more thorough theoretical perspective, [14] discussed Management Information Systems (MIS) and their potential to improve institutional efficiency, data consistency, and decision-making, one way to conceptualize the Romero [15]. The Mendoza system is a web-based, specialized MIS created for specific academic objectives. The same is true of RSSMS, which streamlines research documentation and student-research adviser communication by combining monitoring and support services into a single platform.

Furthermore, [16] underlined that achieving quantifiable operational efficiency across institutional processes is the aim of deploying such MIS solutions. The design goals of RSSMS align with the functional and architectural foundations of earlier studies, aiming to centralize and modernize research workflows through automation and digital integration.

Institutional Challenges and the Need for Centralized Support Systems

The continuous academic and institutional difficulties that students and Higher Education Institutions (HEIs) face underscore the necessity for centralized digital support platforms. Research in [17] shows that financial, administrative, and overall academic stress significantly affect students' performance and general well-being. By offering a centralized research support structure that reduces administrative burdens and improves accessibility to institutional resources, the Research Support Services and Monitoring System for Saint Michael College of Caraga (RSSMS) helps achieve this goal and encourages student success and retention.

User acceptance and perception are critical to effectively implementing centralized platforms in educational institutions. [18] asserts that perceived utility and usability significantly impact the efficacy of academic e-governance systems. This realization is crucial for the design of RSSMS, as responsive features and user-friendly interfaces will encourage faculty and student involvement. By incorporating these usability guidelines into RSSMS, the system ensures that users will actively participate and trust it. While also automating research procedures.

From an institutional standpoint, HEIs are under increasing pressure to maintain their adaptability and resilience amid budgetary constraints and technological demands. Adopting efficient, automated systems that can adapt to rapid institutional changes is necessary for organizational resilience, as discussed in [19]. Using this framework, RSSMS is a tool that improves operational efficiency in research management by centralizing monitoring, tracking, and service coordination. This kind of integration encourages flexibility and institutional preparedness for changing academic demands.

Additionally, [20] underlined that a key factor influencing students' adoption of academic technologies is their impression of institutional support. Students' engagement and success rates skyrocket when they feel their school is trying to provide organized, easily accessible learning resources. RSSMS created a unified digital research platform to show institutional dedication to assisting research activities through easily accessible services, methodical monitoring, and open lines of communication. RSSMS created a unified digital research platform to show institutional dedication to assisting research activities through easily accessible services, methodical monitoring, and open lines of communication.

The Role of System Transparency and Real-Time Status Tracking

Modern academic and administrative information systems must provide transparency and real-time tracking, impacting user satisfaction, accountability, and productivity. Real-time progress visualization and dashboard designs are crucial for inspiring students, especially in capstone and research projects, according to studies in [21]. Throughout the research process, these interactive dashboards encourage self-regulation and engagement by giving users a clear sense of progress and task completion. Systems like the Research Support Services and Monitoring System (RSSMS) at Saint Michael College of Caraga, which can track across multiple RSSs to boost productivity and user awareness, need to include these real-time monitoring tools.

As shown in [22], offering instant updates and clear communication channels increases user compliance and reduces redundant queries. The findings show how well-defined tracking indicators and automated alerts improve student conduct and punctuality. By adding these features to RSSMS, faculty and students can track the status of research-related requests in real time, including statistical analysis, ethics review, and grammar checking. This approach promotes easier departmental coordination and accountability.

Additionally, research in [23] highlights that transparency is not only a technical feature but also a psychological necessity in academic settings. Transparent systems reduce uncertainty, foster institutional trust, and ease student anxiety during key academic processes such as approval tracking and research proposal reviews. By giving users real-time updates on the status of their submissions (such as "Under Review," "Approved," or "For Revision"), RSSMS operationalizes this principle of transparency and ensures that students remain informed and confident throughout the process.

Research in [24] supports these findings by highlighting the importance of user experience (UX) design in preserving system efficacy and clarity. Clear progress indicators, visual cues, and an intuitive user interface are essential to any successful academic service platform. By implementing these UX principles, RSSMS users can easily interpret system feedback, quickly understand their research progress, and engage in a smooth interaction that meets contemporary academic digital service standards.

Multi-Role Management and Workflow Automation in Academic Systems

Academic systems need workflow automation and multi-role management to address issues arising from manual administrative processes. The research in [25] showed that university operations became more efficient after implementing automated workflow systems, which reduced processing time by 40% by eliminating manual routing and reassignment tasks. The research shows that automation technology boosts institutional performance by simplifying recurring operations, which match the RSSMS's core purpose at Saint Michael College of Caraga. The RSSMS can process research-related requests through automated routing to achieve systematic service transaction handling with minimal delays.

Effective role-based management is crucial for maintaining security, ensuring structured task distribution in educational systems, and streamlining workflows. Research in [14] emphasized the importance of Role-Based Access Control (RBAC) in platform design to prevent unwanted access to sensitive academic data and to support logical workflow progression. Applying this principle to the RSSMS ensures that research requests are automatically directed to the right people only—for example, a request marked as "For Statistical Review" would be visible only to the statistician—protecting operational integrity and confidentiality.

The research by [26] examined RBAC deployment methods in Philippine universities to demonstrate that the level of detail required in access management systems must be for academic organizations with complex organizational structures. The research findings align with RSSMS's organizational needs, as it operates with research coordinators, grammarians, and ethics reviewers. Implementing RBAC principles at the local level enables staff members to work within established boundaries that support efficient teamwork across departments.

To support these conclusions, [19] examined automated document routing systems and verified that automated role-based assignment is the most reliable and effective method for handling intricate, high-volume processes that call for advice from several specialists. Research service requests can proceed seamlessly from submission to final review without the need for human oversight by integrating this model into the RSSMS framework. This automation strengthens accountability, reduces administrative bottlenecks, and improves service reliability across all research-related transactions.

1.3 Objective of the Study

The study is to create and deploy the Research Support Services and Monitoring System (RSSMS) to enhance research service management at SMCC by providing an organized, transparent, and efficient workflow for assigning, processing, monitoring, and documenting research-related services.

Specifically, it aims:

1. To develop a web-based system that automates the student's submission and checking process for research support documents, specifically for Grammarly & AI Checking, Human Grammarian, Statistical, Ethical submission, and References validation.
2. To design and implement a monitoring system for tracking the progress and status of research support document validation.
3. To evaluate the systems functionality, efficiency, and usability, and efficiency through ISO/IEC 25010 standards to ensure its effectiveness in addressing the identified problems.

1.4 Scope and Limitations

The Research Support Services and Monitoring System (RSSMS) is a digital platform enabling Saint Michael College of Caraga (SMCC) to automate research transactions through an efficient management system. The system replaces the disorganized research proposal, request handling, monitoring activities, and documentation processes. The system integrates all research processes into a single platform, enabling students, advisers, and research personnel to communicate through a single interface while tracking real-time activities.

The system operates as a restricted platform that grants access to official research participants, including students and their faculty advisers, and research staff members, including the librarian, statistician, human grammarian, and ethics reviewer. The Research Director or authorized research staff member holds the single superuser account, which grants them complete system administrative control. The administrator performs all user account management tasks, assigns personnel to research projects, and tracks progress and reviews submissions to create statistical reports for institutional use. The system enables administrators to perform user management tasks and staff assignments, track research progress, and create reports. The system enables administrators to track user activities through its notification system, which provides immediate process updates and maintains complete transaction transparency.

The system enables students and faculty researchers to access their personal accounts for document uploading and service requests, including grammar, plagiarism checks, statistical analysis, ethics validation, library resource access, and real-time submission tracking. The system sends student requests and documents to their designated personnel after they complete the submission process. The system enables personnel to review submissions and provide feedback before approving or rejecting them in accordance with institutional research guidelines.

All administrative functions are restricted to the superuser or administrator, while end-users are limited to project-related activities and assigned transactions. The system also supports anonymous communication between students and personnel, enabling private clarification and feedback within the platform

The RSSMS operates within the institution's network and internal database, ensuring all processes and data remain secure and confined within SMCC. It addresses the institution's internal research management requirements, focusing on document submission, transaction tracking, progress monitoring, and communication. For advanced functionality, the system integrates secure external API connections for plagiarism detection and grammar checking. These tools allow uploaded research documents to be compared against global databases to verify originality and assess grammar accuracy.

Aside from this API integration, the RSSMS does not connect to external research publication platforms or academic repositories. All research data, files, and communications are stored securely in the institutional database and are accessible only by authorized personnel.

Through this scope, the RSSMS ensures a controlled, confidential, and efficient environment that supports the end-to-end management of research services at Saint Michael College of Caraga.

1.5. Definition of Terms

RSSMS authentication — ensures that users are who they say they are by checking their login information. Using RSSMS features keeps users safe.

Database — A structured set of data that saves all the information about a system, like user accounts, submissions, feedback, transactions, and messages between students and staff. MongoDB is usually used to handle databases.

Gcash Payment Gateway — integrates with RSSMS, allowing users to pay for research-related services online safely and efficiently.

Higher Education Institutions — want to help students get degrees, strive for greatness, and reach their full potential.

Role - Based Access Control — is a type of security that limits system access based on the job role of a user within a company rather than the user themselves.

Research Support Services and Monitoring System (RSSMS) — is a web-based tool that makes it easier for students to submit, review, and monitor their research results. It lets students talk to each other and gives jobs to human grammarians, statisticians, librarians, ethics officers, and Grammarly & AI Checking services.

Research Data Management (RDM) - An organized or integrated system/practice for controlling, preserving and using research data over its lifecycle that enables data to be used regardless of the times and lengths of its production.

Digital Ecosystem - A connected digital environment consisting of digital tools and platforms which are used for academic work, research, communication and data management. It integrates physical and digital models for total support services.

Information and Communication Technology (ICT)-Based Systems - Are those systems that use technological infrastructure (i.e., software, networks, and digital platforms) to facilitate management, monitoring and provision of institutional services in general, predominantly research.

CHAPTER 2

METHODOLOGY

2.1 Research Design

The Research Support Services and Monitoring System (RSSMS) for Saint Michael College of Caraga uses the Design and Development Research (DDR) method. This approach allows the creation of new systems that address the real needs of institutions. The system manages student submissions, administrative tasks, and employee evaluations before providing final approvals and certifications. As a comprehensive research support automation platform, RSSMS underwent multiple rounds of design refinement based on stakeholder feedback. This process ensured optimal performance, a user-friendly interface, and compliance with institutional research management standards.

2.2 Software Development Life Cycle

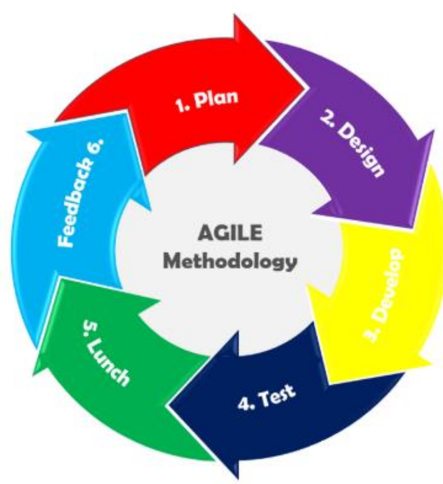

Figure 1. Agile Methodology

Figure 1 shows the Agile Software Development Life Cycle, which is made possible by the Research Support Services and Monitoring System (RSSMS) development method. Because it allows for flexible development, ongoing stakeholder involvement, and small steps forward, the agile development method is perfect for bringing students, administrators, and research staff together in real time.

The RSSMS development process used several sprints to make modules that worked, beginning with student submission and continuing with administrative routing, personnel review, feedback notification, and certification issuance. During planning sessions at the start of each sprint for research operations at the school, the Research Office worked with librarians, statisticians, and student representatives to ensure system features were correct. After each sprint, the researchers kept trying the system to improve how the platform and interface worked while keeping user and institutional needs in mind.

2.3 System Architecture

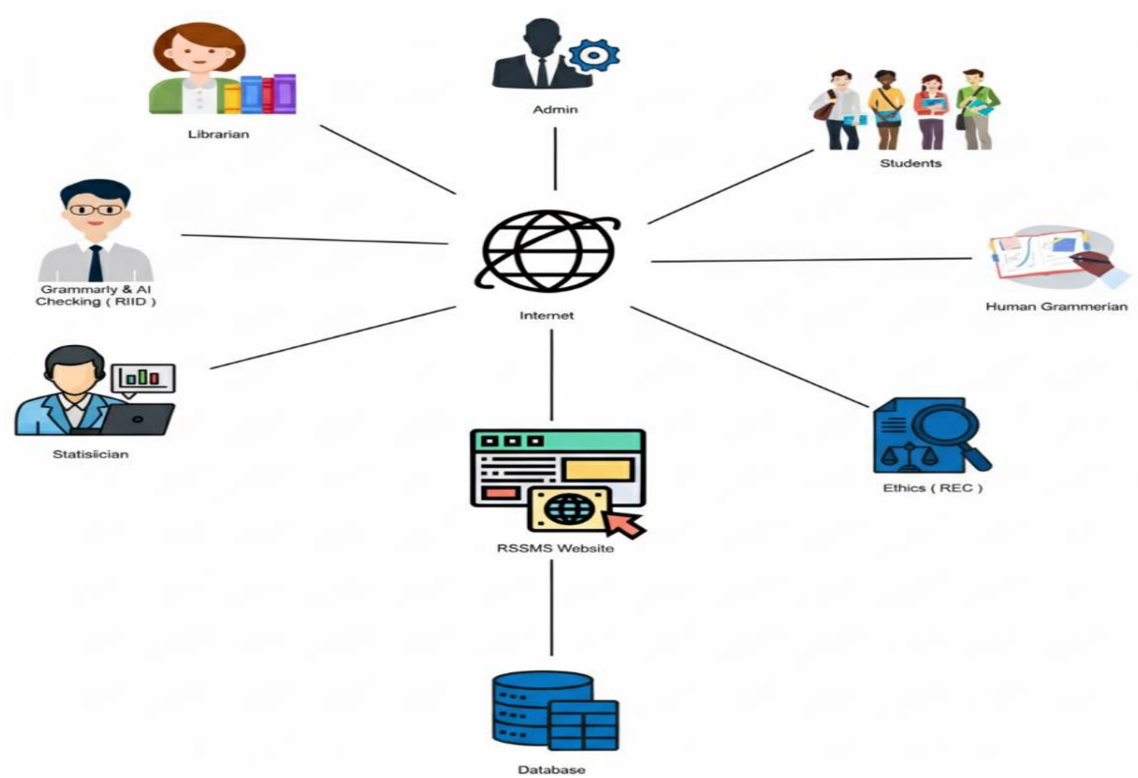

Figure 2. System Architecture

As shown in Figure 2, the RSSMS architecture is a web-based, centralized platform that links all system players (Students, Admin, Librarian, Human Grammarian, Statistician, Grammar & AI Checking, and Ethics (REC)) through the Internet. Everyone can use the RSSMS website to send documents, complete tasks, control services, and talk to each other.

The admin is in charge of user accounts, services, and transactions. Staff members in charge of certain services get student entries and process results. The architecture of the system stores and manages all exchanges through a central database. Users' information, study files, system logs, payment information, assessment feedback, and certification documents can all be stored safely in the database.

The database connection to the RSSMS website allows for the recording of all system actions while fully following the proper steps. The system's architecture allows safe and effective work processes that help the institution study, provide services, and keep an eye on things through organized and clear operations.

Students must pay for the Grammarly & AI Checking service. When a student adds a file and pays with GCash, the system sends the file to the right person for review. The student can access the result file and certificate once the submission is accepted. On the other hand, students can submit their files without an initial payment for services such as Statistician, Librarian, Human Grammarian, and Ethics. Payments are only required after the personnel have checked and approved the submission, ensuring that students pay only for verified and completed services.

2.4 Conceptual Framework

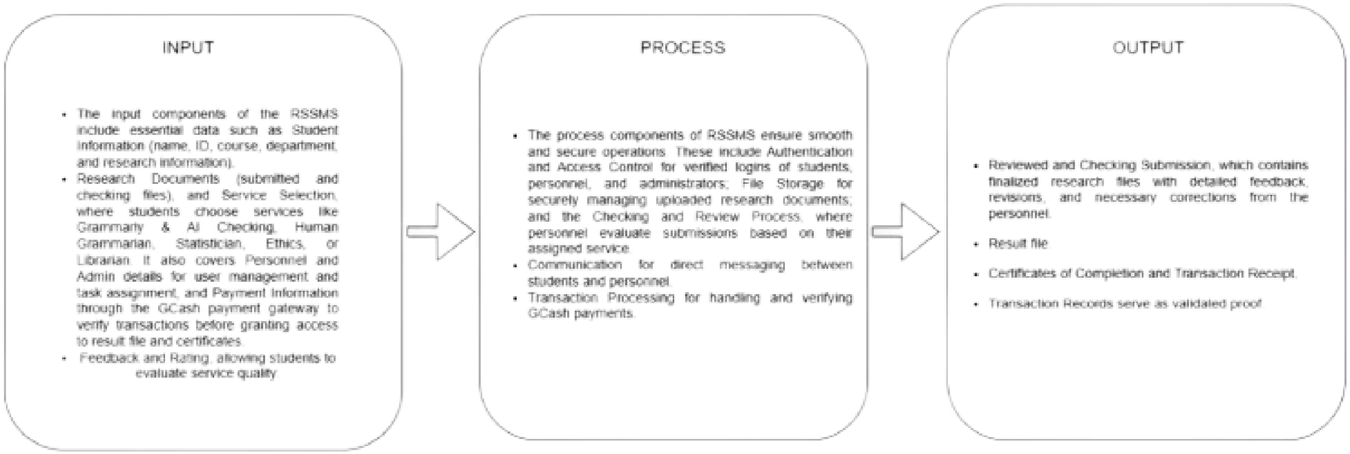

Figure 3. Conceptual Framework

This Diagram illustrates how RSSMS functions. RSSMS serves as a primary tool connecting students with all support staff, ensuring that services related to research run smoothly. The services are easy for students to access and utilize. They can turn in their study files and see how their submissions and state are going. They will have to wait three to five work days after that. Through the RSSMS, students can talk to the staff members who are responsible for them. There are also ways for them to rate and comment on services, download shared results and awards, and pay their transaction bills online.

This process makes it easy for students to study, boosts communication, and clarifies things. The RSSMS gives different types of people different administrative jobs. These include admins, students, statisticians, librarians, human grammarians, ethics, grammar, and AI checking staff. Each job has specific responsibilities that ensure the work gets done quickly and well.

The administrator creates user accounts, assigns services, manages departments, checks submissions, and keeps track of transactions and comments. Grammar and AI Checkers use outside software to do automatic checks. Statisticians analyze data, librarians check references, human grammarians check grammar by hand, and ethics staff check for ethics violations. Each job includes checking student files, giving feedback, keeping statuses up to date, and giving out completion certificates.

2.5 Use Case Diagram

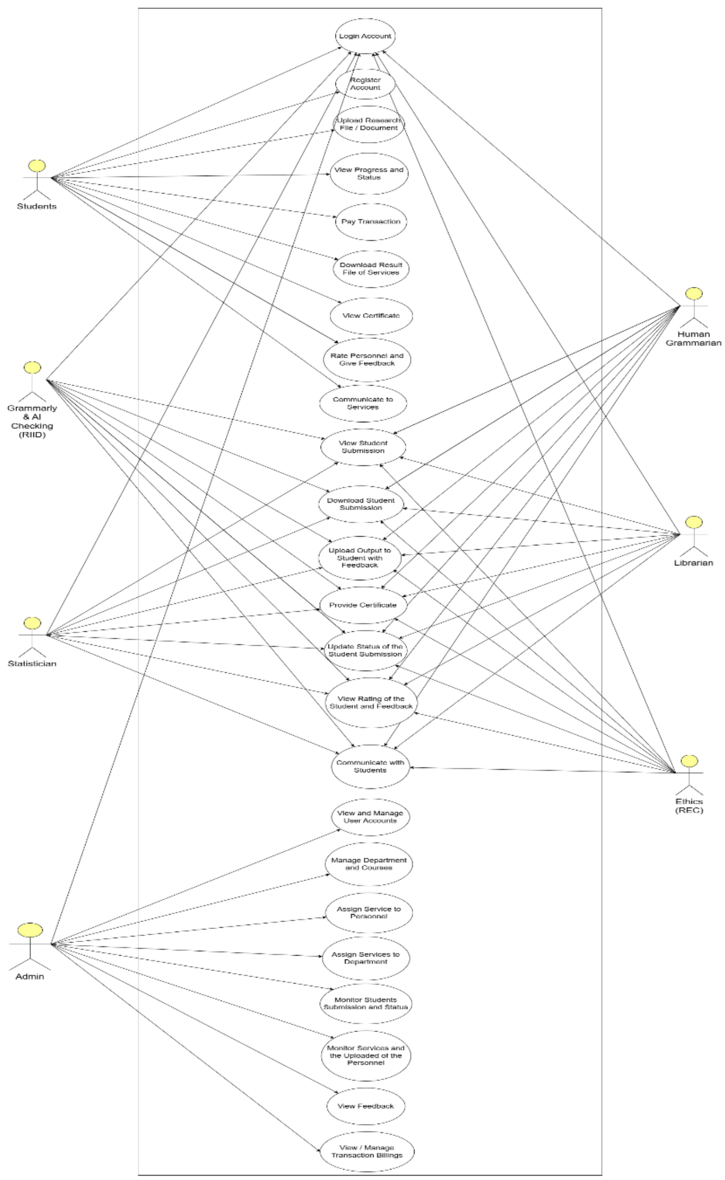

Figure 4. Use Case Diagram

The operational flow for managing user accounts and services in RSSMS features a visual representation in Figure 5. After logging into RSSMS, the administrator accesses a centralized dashboard to manage various system components. The system allows administrators to create and update user accounts, assign service personnel, designate department personnel, and manage services to ensure appropriate distribution

Explains how the RSSMS interacts with its many users by emphasizing its features and goals. The primary users who choose services, upload research files, monitor progress, communicate with staff, and get results are students. Additionally, students can use the Internet to pay fees, review staff, and download certificates. This collection of use cases guarantees that students can complete their research requirements on a single platform, resulting in an open and effective process from start to finish.

The RSSMS includes several support positions: administrator, statistician, librarian, human grammarian, ethics, and Grammarly & AI. Verifying the staff, each of whom has specific duties. The administrator manages user accounts, service distributions, departments, and transactions for smooth operation. Other staff members examine students' submissions based on their areas of expertise.

Statisticians handle data analysis, librarians check references, human grammarians check language, ethics examines ethical adherence, and automated checking is done by grammar and AI staff using external software. Viewing and downloading submissions, providing feedback, updating status, and awarding certificates constitute the work. This structured interaction ensures accountability, proper workflow division, and consistent support for students' research projects.

2.4 Activity Diagram

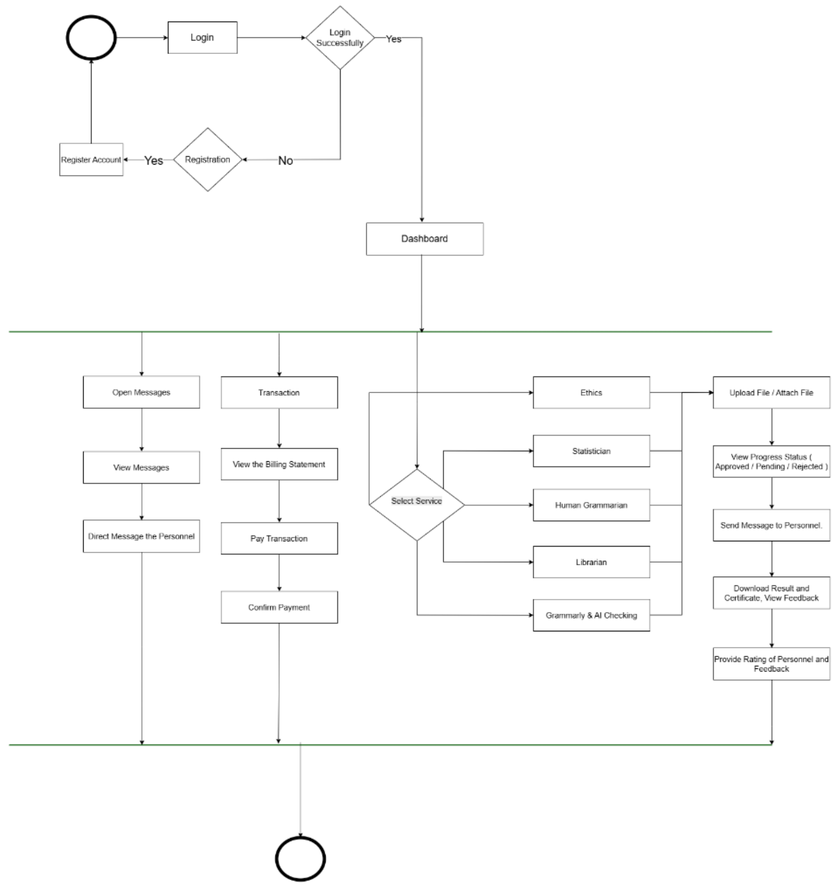

Figure 5. Activity Diagram - Students

Demonstrates the RSSMS system, which begins with student login and then allows service selection between grammar and AI checking, librarian reference, statistical, and ethics reviews, and a human grammar checker for manual grammar reviews. The student chooses the service they need before submitting their research paper to RSSMS for format validation and submission approval.

The RSSMS verifies the document before sending it to the appropriate reviewer for assignment, while students can monitor their submission status in real time through the system.

The system maintains an efficient workflow for academic research document processing from submission to review through its structured process.

The student receives access to review feedback, and a reviewed document is downloaded with a completion certificate after the review process finishes. The student can use the system to contact personnel for additional edits or clarification needs. The student reviews the payment details before making an online payment to ensure a smooth transaction. The student can rate personnel and provide feedback before completing the activity flow. The activity diagram enables students to deliver an efficient interactive research assistance system.

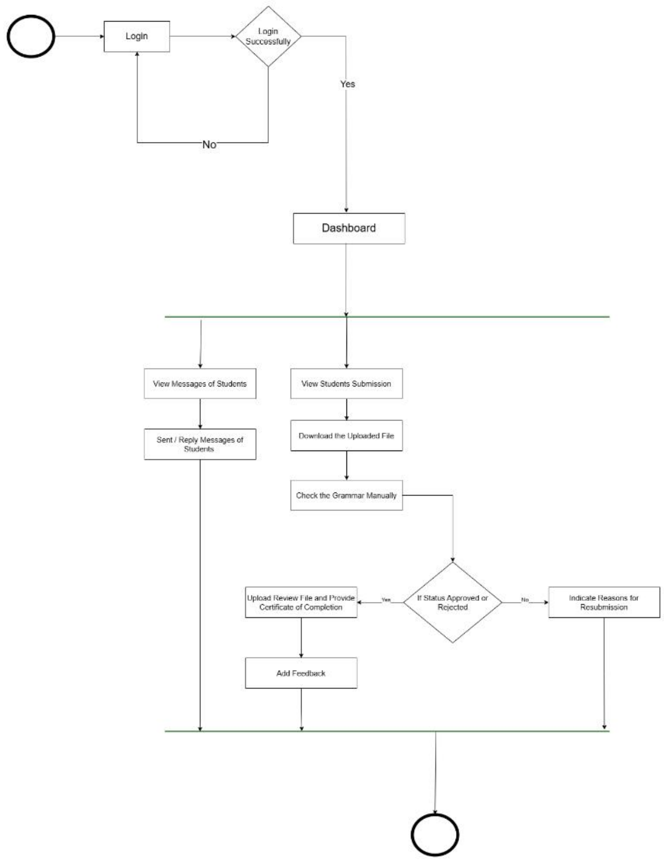

Figure 6 Activity Diagram – Human Grammarian

This figure illustrates the step-by-step manual research paper assessment process, which follows a structured format for designated personnel. The Human Grammarian service process starts when students select their service and then upload their research document before submitting it for review.

The RSSMS sends an instant email notification to designated grammar personnel after students submit their files for review. The grammarian accesses the system through login credentials to download the file for manual evaluation of grammar, language clarity, tone, and overall document coherence. The grammarian reviews the file before issuing a completion certificate or returning it with revision comments to the student.

The system sends the results to students through email notifications. Students can access their approved documents through download, but must resubmit their work after revisions. The messaging system enables students to maintain direct contact with the grammarian throughout their work process. Students assess the service quality through feedback, which personnel review to enhance service delivery and maintain high standards.

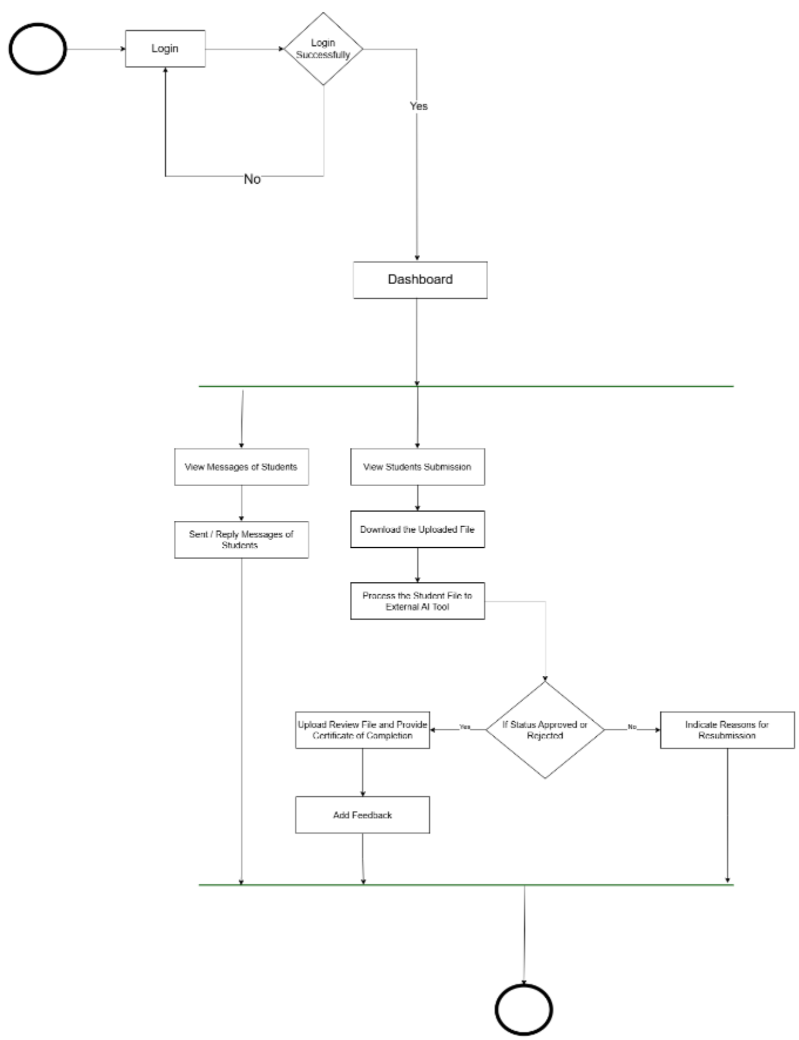

Figure 7. Activity Diagram – Grammarian & Ai Checking

The Diagram shows how email notifications are built into the system, ensuring that students and service personnel receive real-time alerts about submission events. When a file is submitted, the system automatically emails the personnel who can view the submission from their dashboard.

The system routes the file to an AI tool for processing and feedback. After the AI finishes the personnel, approve the submission by attaching or rejecting the certificate, prompting a resubmission. As soon as a status flips, the system sends an email to the student; the message either confirms that the service is complete and the file and certificate are ready for download, or it includes a rejection note outlining the required changes and asking for a resubmission. Staff members can also correspond with students. Reply to their messages. In short, this arrangement delivers an efficient real-time notification pipeline that enhances the user experience and keeps all parties informed and actively engaged throughout the transaction.

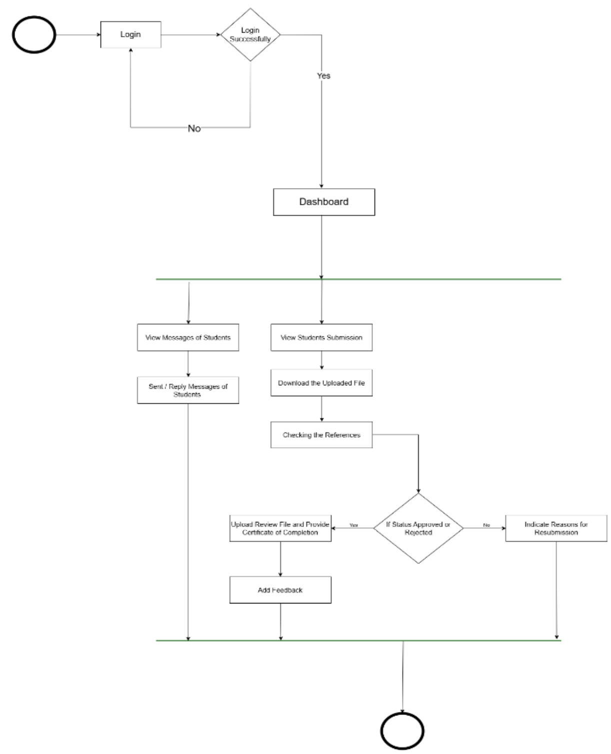

Figure 8. Activity Diagram – Librarian

The RSSMS includes a diagram in Figure 8 that shows the structured process for handling student research papers during reference verification and checking. The RSSMS accepts research papers from students who select a librarian service, store the documents, and send an email notification to their chosen librarian.

The librarian accesses RSSMS to check pending submissions and download the documents after logging in. The checking process involves verifying that citations match the correct sources and references follow the required formatting and citation style guidelines. The system verifies that the research paper meets the institution's academic referencing standards. The librarian completes the evaluation process by uploading a completed version of the submission, including a certificate of completion, or by rejecting it with feedback for another attempt.

The RSSMS sends email notifications about submission results to students who can take the necessary steps. Students can contact librarians directly for clarification to maintain smooth coordination during the review process. Students provide their assessment of the service through ratings and feedback, which librarians can access to improve service quality. The system maintains complete transparency and accuracy and ensures accountability through its structured academic reference and citation management workflow.

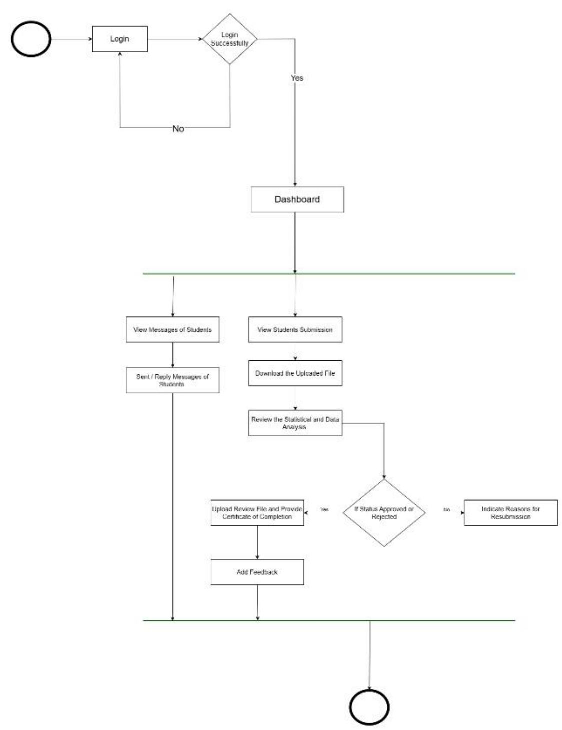

Figure 9. Activity Diagram – Statistician

The RSSMS diagram outlines the steps in processing student research submissions for statistical analysis and verification. After students choose their statistician service and upload their research, the system notifies the assigned statistician via email, ensuring document security. Before downloading the available documents, the statistician logs into the system using their credentials to process any pending submissions.

Three steps make up the review process: evaluating the completeness of the data, validating statistical techniques, verifying statistical conclusions, and providing readability explanations. The review process ensures both methodological soundness and reasonable research outcomes. After reviewing the file, the statistician either rejects it with comments and revision instructions or approves it for final review and upload of statistical results and completion certificate.

Students receive automated email notifications from the system about the results of their reviews. In addition to having the option to resubmit their work after revisions, students can use the system to access their approved files. Students can interact with statisticians via the RSSMS messaging system to get clarification and feedback on their work. After reviewing the service, students provide feedback that statisticians can use to improve its quality. The systematic review process ensures accurate results, complete transparency, and accountability in statistical validation and review operations.

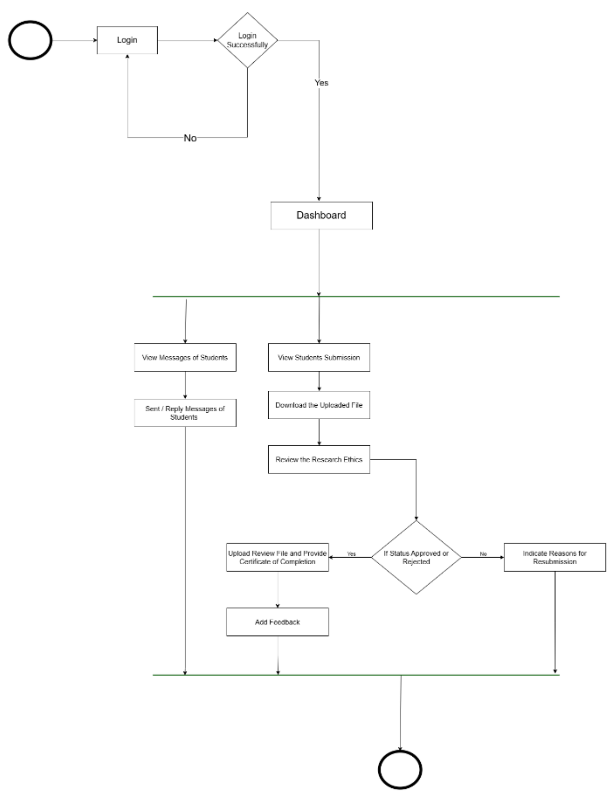

Figure 10. Activity Diagram – Ethics (REC)

The RSSMS uses Figure 7 to show its systematic approach for ethical research paper review within the system. The RSSMS receives research papers from students who select ethics as their service, while the system automatically sends notification emails to Ethics personnel. The personnel access the RSSMS system to review submitted documents for ethical compliance with institutional rules, research integrity standards, consent procedures, and confidentiality requirements.

The review process verifies that research meets all ethical standards before advancing to publication or defense stages. The Ethics personnel complete their review of the document by issuing an approval certificate or providing detailed editing feedback for denial. The RSSMS delivers the review decision to students via email. The student can retrieve the document and certificate after downloading or resubmitting the document for editing. The messaging feature enables both parties to exchange messages to understand better and maintain transparency. The student provides service feedback and ratings to maintain high standards for ethics review services.

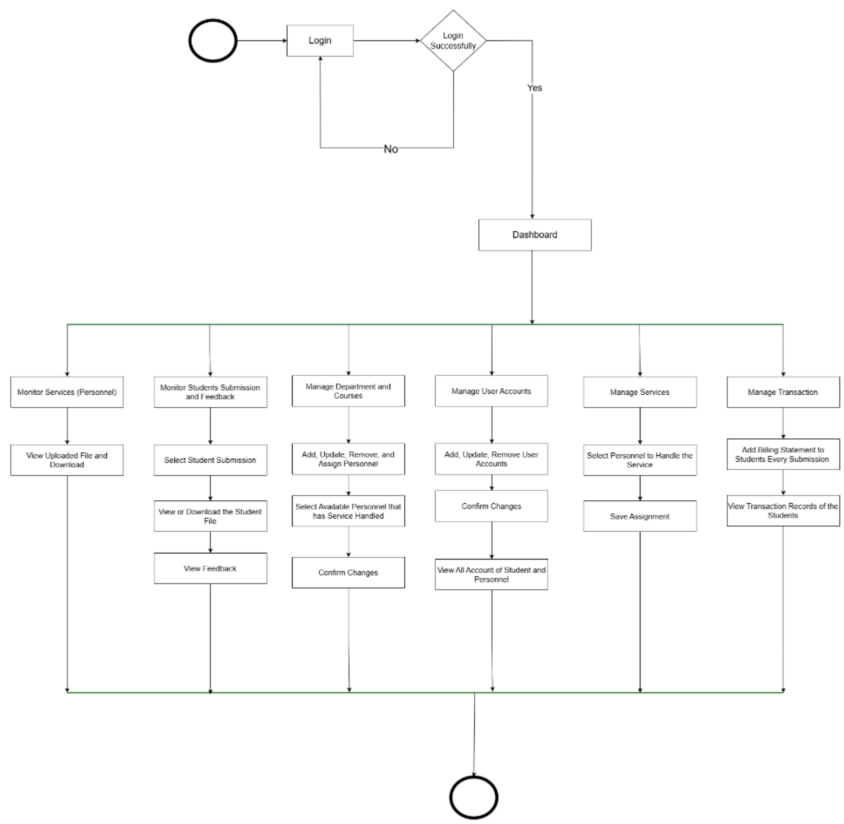

Figure 11. Activity Diagram – Admin

The operational flow for managing user accounts and services in RSSMS is displayed. After logging into RSSMS, the admin accesses a centralized dashboard to manage various system components. The system enables administrators to create and update user accounts, assign personnel to services and departments, and oversee services to ensure proper distribution.

The system enables administrators to track student submission status across all services while maintaining workflow efficiency and providing access to personnel and student file upload. The feedback module provides admins with performance data on service providers, helping them solve problems and enhance service quality. The system administrator maintains control over all system operations through their ability to review billing statements and transaction records.

The platform enables instant issue resolution through its built-in communication system, which connects personnel with students. The system maintains organized operations that enable administrative control, transparency, and coordination among all system components to support efficient interactions between student service personnel.

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2.7 Sequence Diagram

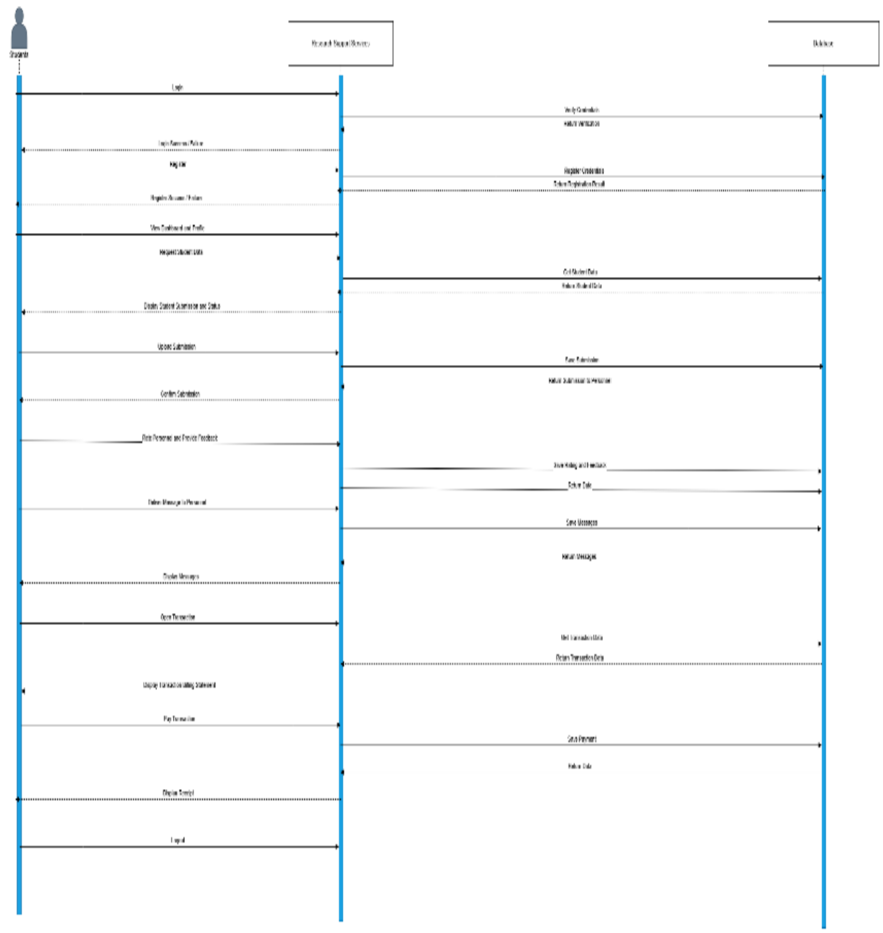


Figure 12 Sequence Diagram – Students

The process of student-service personnel interaction at RSSMS appears to demonstrate the steps of communication for document submission and review. Students access the RSSMS dashboard to select their preferred service before uploading their research document. The RSSMS system sends automatic email notifications to personnel after file storage, while students receive confirmation of their successful document submission.

The system enables students to monitor request status and directly communicate with personnel for clarification, creating transparency and immediate user interaction. After finishing their review work, the personnel update the review status in RSSMS and send an email notification to the student.

Students access the system to check their reviewed files and completion certificates or make revisions before resubmitting their work if needed. The GCASH payment gateway enables students to make payments, resulting in a smooth and seamless process for accessing their results and certificates. Students need to finish their payment before they can access their result file and certificate. Students provide service ratings and feedback after completing their work, which helps the service improve continuously. The RSSMS uses a structured process to maintain clear communication paths, track progress, and deliver services efficiently.

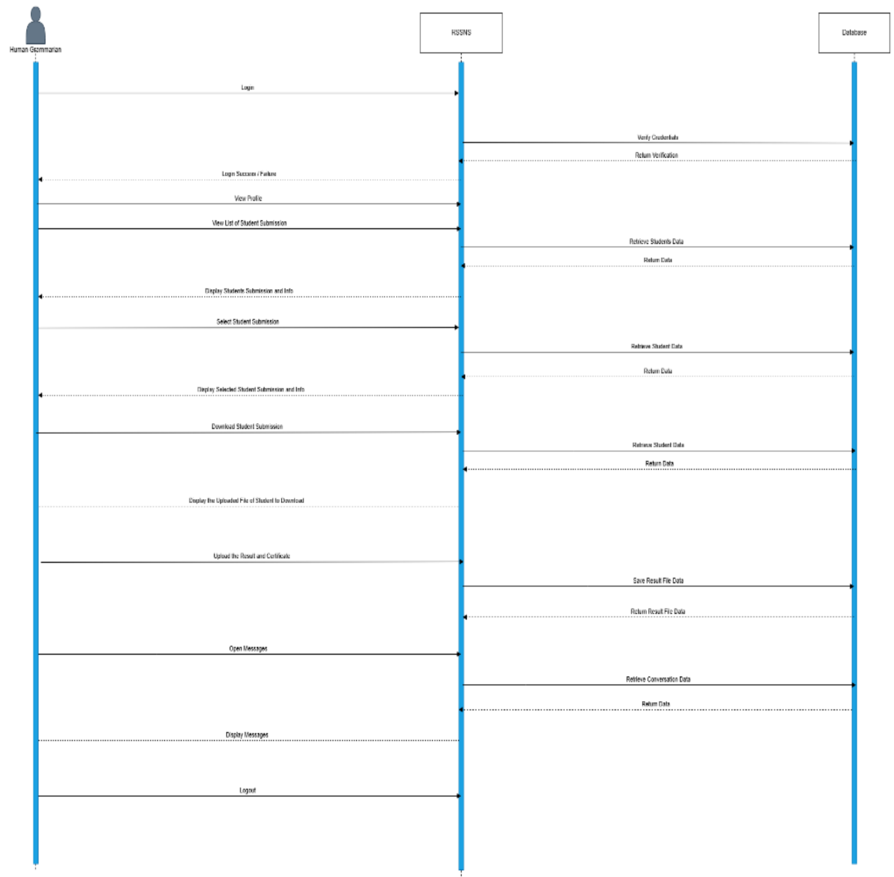
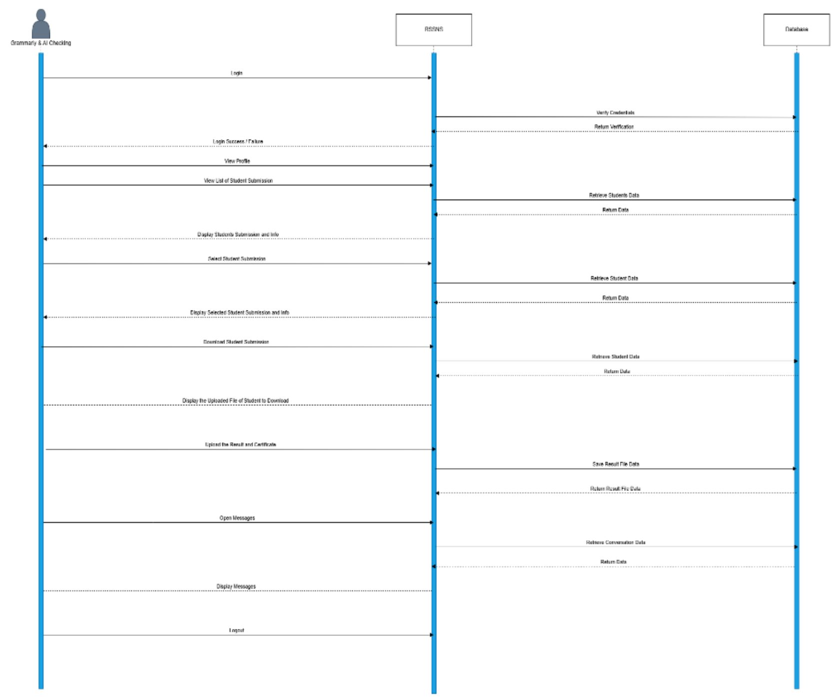


Figure 13. Sequence Diagram – Human Grammarian

The student starts by accessing RSSMS and reaching the human grammarian services, where they upload their document to assess its grammatical and writing quality. The RSSMS generates an email alert to the designated human grammarian personnel who can begin evaluating the submission.

The grammarian begins their work by downloading the student submission file for manual evaluation of grammatical mistakes, sentence organization, pragmatic elements, and clear writing. The human grammarian finishes their review by uploading the corrected file and marking the submission status as Approved, rejected with revision needs, or resubmitted.

The RSSMS sends email alerts to students about review results and output, while students must complete payment for approved submissions to access their reviewed files and certificates. The system grants students access to their reviewed file and certificate after verifying their



payment transaction through RSSMS. The system provides students and grammar personnel with a communication tool to request clarification.

Figure 14. Sequence Diagram – Grammarian & Ai Checking

The RSSMS diagram shows how students' research submissions undergo automated and semi-manual checks for grammar and AI detection. The system allows personnel to access their dashboard, where they can upload files that trigger automatic email alerts to Grammarly and AI checking personnel.

This process makes it easy for students to study, boosts communication, and clarifies things. The RSSMS gives different types of people different administrative jobs. These include admins, students, statisticians, librarians, human grammarians, ethics, grammar, and AI checking staff. Each job has specific responsibilities that ensure the work gets done quickly and well.

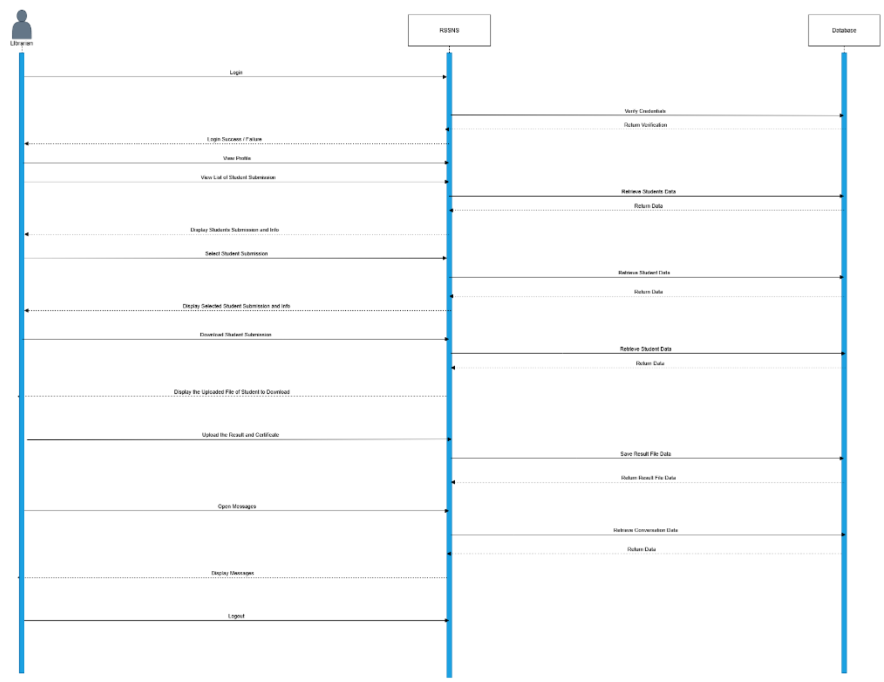


Figure 15. Sequence Diagram - Librarian

When students log in to the library system to share their research, the RSSMS starts its librarian sequence. After uploading their file to the RSSMS, the student sends it for reference review, which sends an email alert to the assigned librarian staff.

The librarian looks at the submission and ensures it meets the school's requirements. These include rules for formatting and citing sources. The librarian finishes the review process by adding the result file and certificate to RSSMS. The message indicates whether the student's work received approval or requires resubmission.

The RSSMS lets students know through emails about the results of their entries, so they can make any changes or resubmit them as needed. After being approved, the system takes students to a page where they can pay before letting them see their study results and certificates. After completing the work, the RSSMS verifies that the university pays the students correctly and grants them access to their study results and certificates. The communication tool lets students and library staff share information easily and discuss other issues.

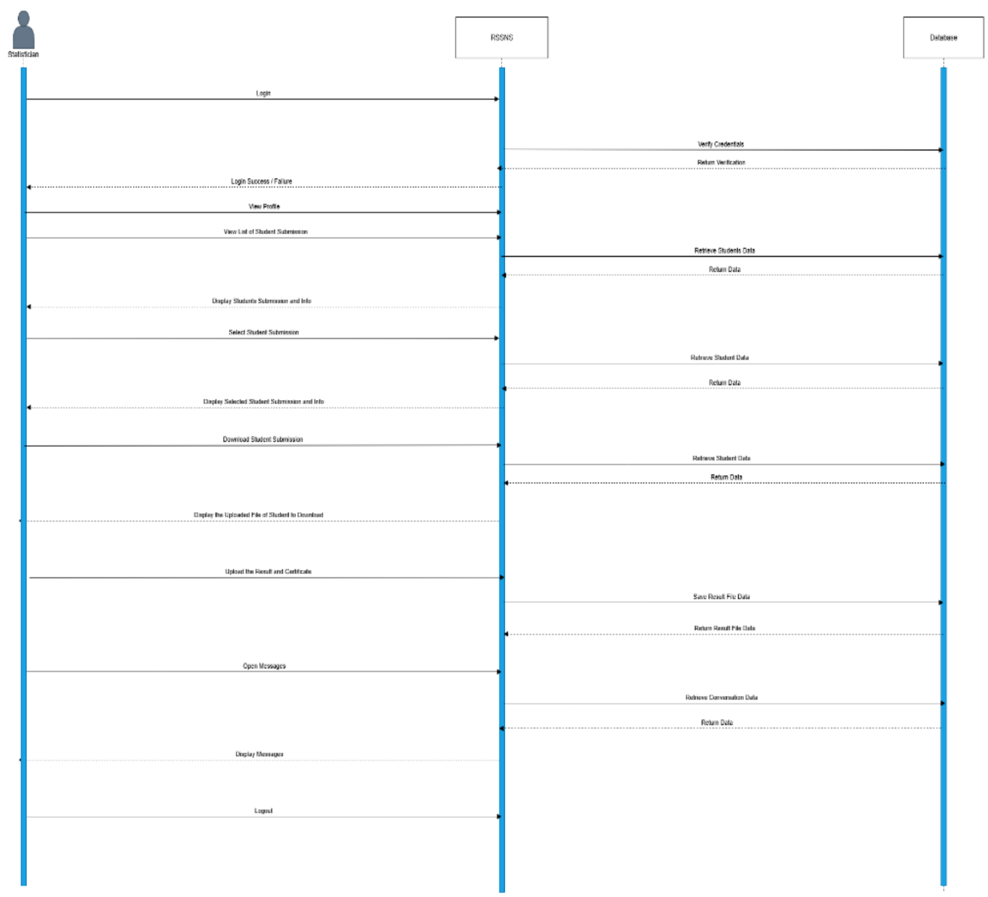


Figure 16. Sequence Diagram – Statistician

The analysts' sequence diagram shows how the organized process of statistical analysis for student data works after students upload their files and assignments to RSSMS. With secure data handling, the RSSMS system lets the statistician know about new entries by email.

Before uploading the results and a completion certificate, the statistician does the necessary statistical analysis and data processing. When a student finishes paying, the RSSMS system changes the state to (Approved, Pending, or Rejected) instead of letting them see their results and certificates.

The system sends email alerts for submissions and readiness results after payment completion. The RSSMS system provides access to the reviewed file and certificate and allows students to resubmit their work in case of rejection. The system enables immediate communication between students and statisticians through real-time clarification, and the rating and feedback system helps maintain service quality and evaluation.

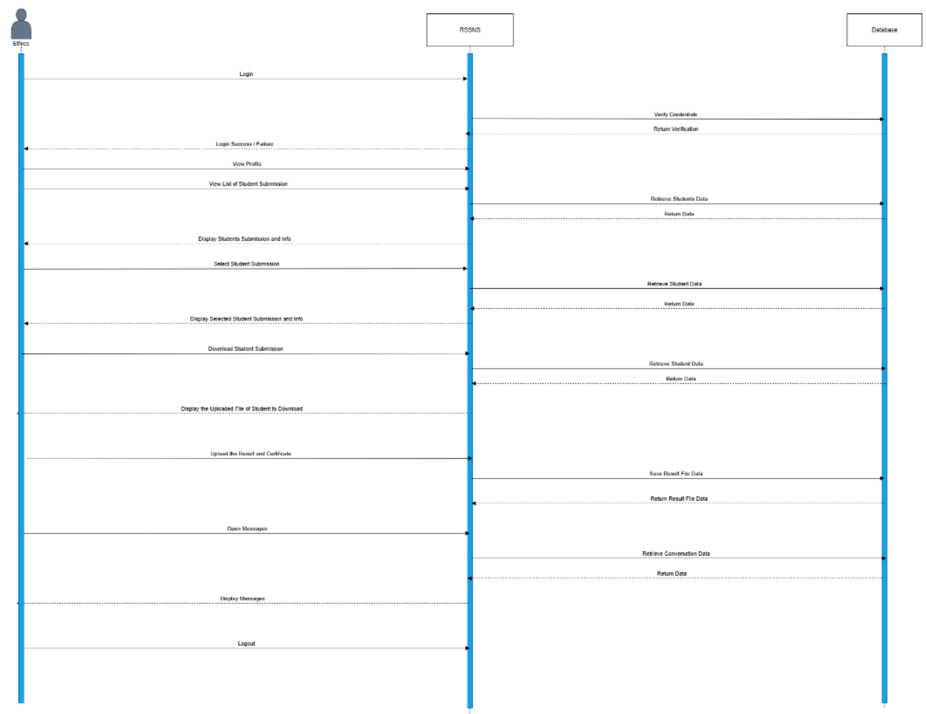


Figure 17. Sequence Diagram – Ethics (REC)

After logging in, the RSSMS starts when students access the Ethics services through the platform. Students must upload their required Ethics documents before submitting their application for review through the RSSMS, which then transfers the submission to the ethics service for review.

The ethics team reviews student submissions to verify ethical compliance before making approval or rejection decisions for the submission. The RSSMS delivers email notifications to students after ethics personnel complete their review, providing submission outcome information and requiring resubmission when necessary. Students who receive approval must complete transaction billing to access their result file and ethics certificate.

After payment verification, the RSSMS system provides students immediate access to their approved submission file and ethics certificate. The ethics service obtains confirmation of process completion for monitoring and reporting purposes through the structured flow, which maintains accountability and ethics compliance, enables real-time communication between student-ethics personnel, and restricts access to sensitive ethics documents in RSSMS.

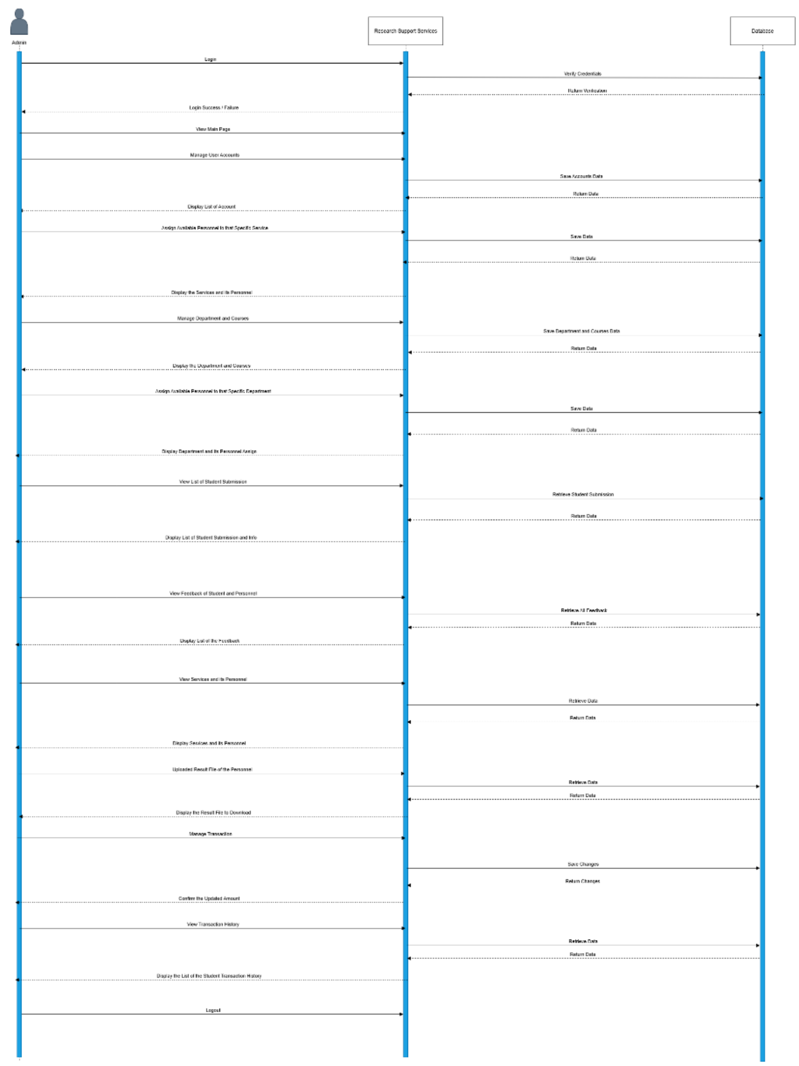


Figure 18. Sequence Diagram – Admin

The flowchart demonstrates the complete sequence of administrative tasks, including Dashboard management, Authentication and Research Support Services, Students and Personnel, and Database operations. The admin starts their process by logging in, which triggers authentication to verify their credentials. The system verifies the admin after authentication so that they can proceed with user account management and department and course administration.

The database processes all administrative changes through add, update, or remove operations, which result in immediate system updates to RSSMS. After completing account, department, and course management tasks, the admin can assign personnel members to research support services, including ethics review, human grammar check, statistical review, and additional research services.

The admin sequence includes student submission tracking, which displays research result files and certificates alongside personnel feedback. The system enables administrators to select feedback from students and personnel, helping them track communication and maintain the review process's efficiency. The Diagram demonstrates how the admin controls research services and personnel oversight through transaction management, submission amount updates, student billing statements, and transaction history access. The administrative structure provides complete management control for RSSMS operations while maintaining system efficiency.

2.7 ERD (Database Design)

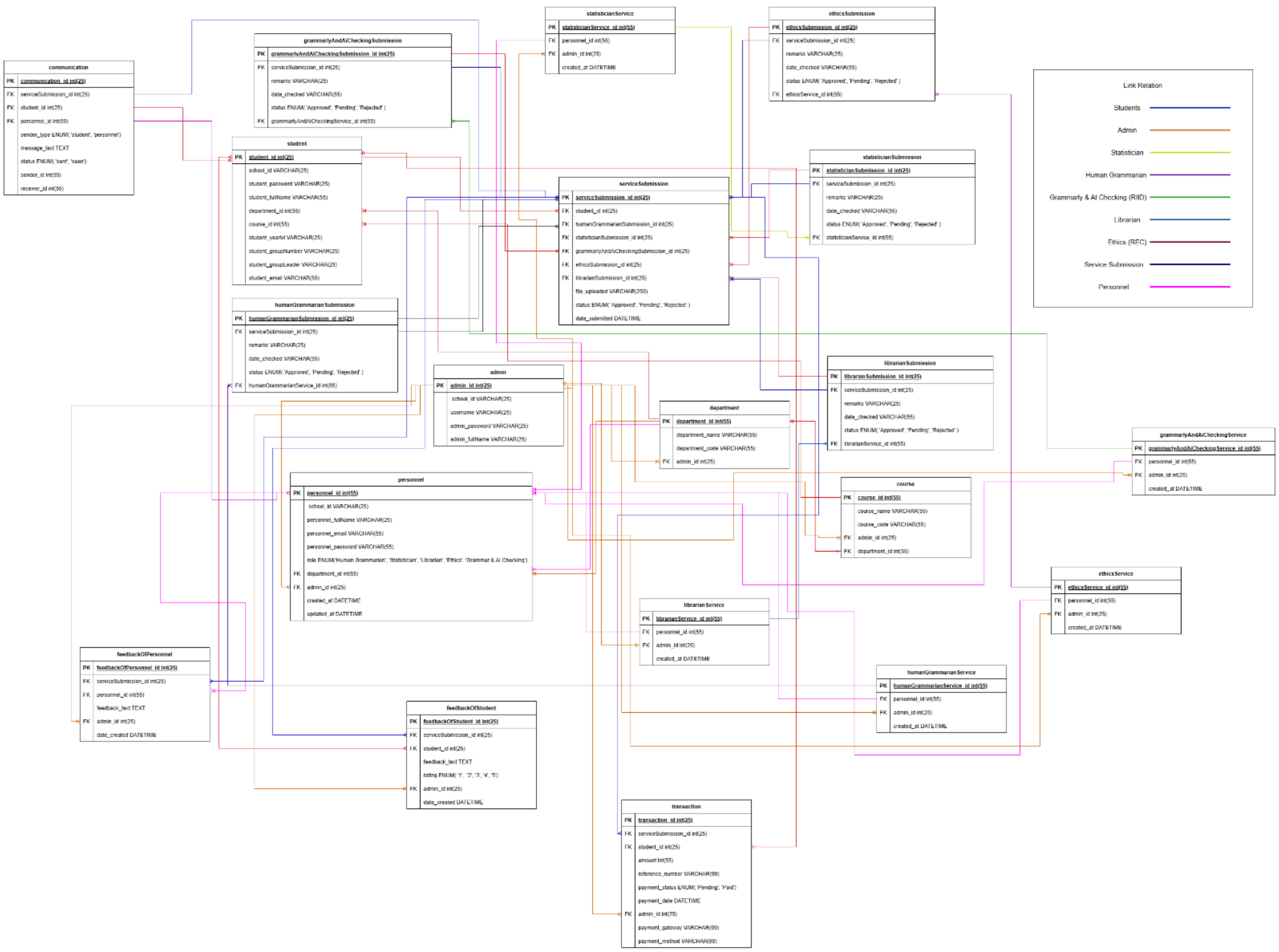


Figure 19. Entity-Relationship Diagram (Database Design)

The database schema demonstrates the complete data structure of RSSMS, which shows how all entities within RSSMS relate to each other. The student entity is the core element of RSSMS because it contains vital student information, including personal details, course and department enrollment, and uploaded submission files. The students maintain a connection to the service submission table, which functions as the central hub for all research submissions.

The submission bridge table connects students to different services via foreign keys that identify service destinations and contains file paths, status information, and submission dates for tracking service personnel handling. The admin controls personnel management through the personnel table while assigning staff members to particular services for handling. The system includes separate tables for each service type (librarian, statistician, human grammarian, grammar & AI checking, and ethics), enabling flexible role management and service submission linking. The admin controls department and course tables to establish organized student and personnel categories, facilitating communication between them.

The system includes a communication table to store all messages that students and personnel exchange with each other. The system enables transaction and billing operations through the transaction feature, which supports GCASH as its payment method. Students must finish payment transactions for service submissions before accessing their result files and certificates. At the same time, the feedback system enables students and personnel to rate services after completion for better accountability and service quality. The system provides secure and scalable data exchange between students and services in RSSMS while maintaining transparency for complex academic support operations.

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2.8 User Interface Design (Prototype)

STUDENT

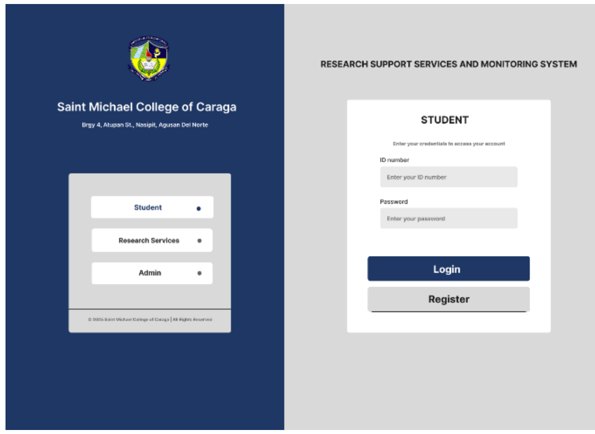

Figure 20. Log in form - Student

Students need to access the RSSMS using their registered credentials or create a new account to register. Securing authentication ensures the RSSMS can accurately track each student's submissions, payments, and service interactions.

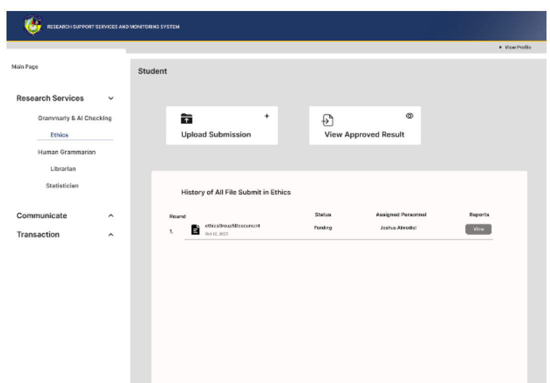

Figure 21. *Student Select Ethics Research Service and Upload Submission*

Grammarly & AI Checking, Ethics, Human Grammarian, Librarian, and Statistician are some service choices students can choose from. Student upload submission for round 1 in Ethics service, then student wait for the approval of personnel. As soon as they share their work, they can see how it did and a record of all the submissions they made for each service. And student will receive an email notification after the submission and the service too.

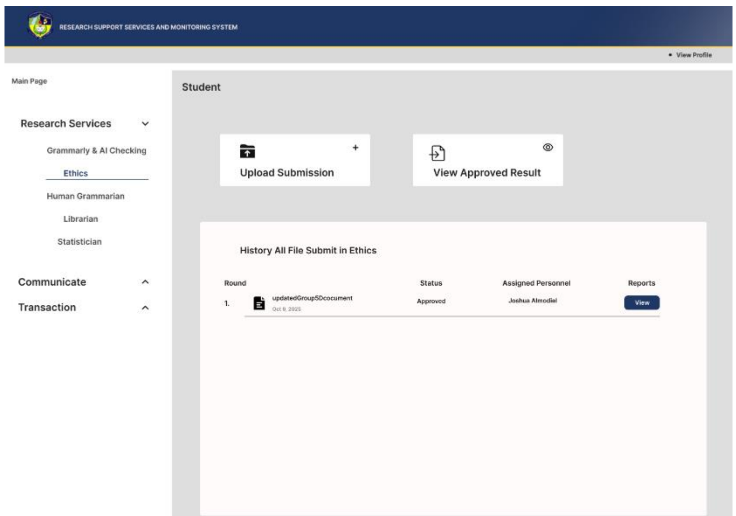

Figure 22. Student Submission Approved in Ethics Service

I The student round 1 submission status is now approved then they can view the approved result, view reports, provide rating with feedback and the certificates of completion. After the review of student submission notify them with email notification that the submission is now approved or rejected.

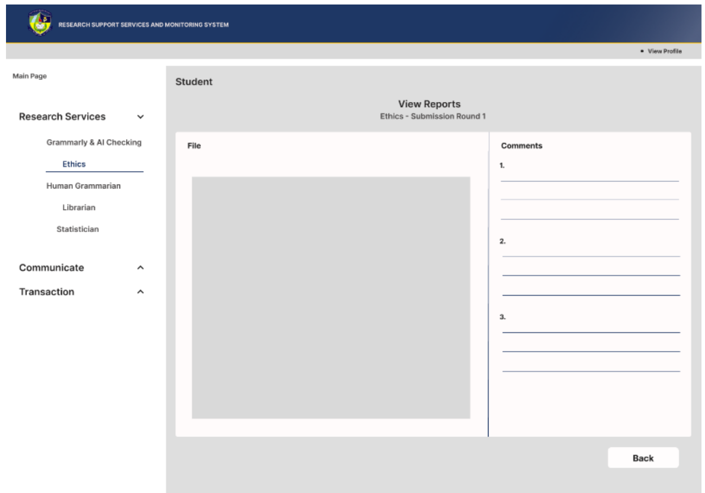

Figure 23. Student Submission View Reports in Ethics Service

The student can view the reports of the ethics service round 1, view file and the comments of the personnel for the student to review.

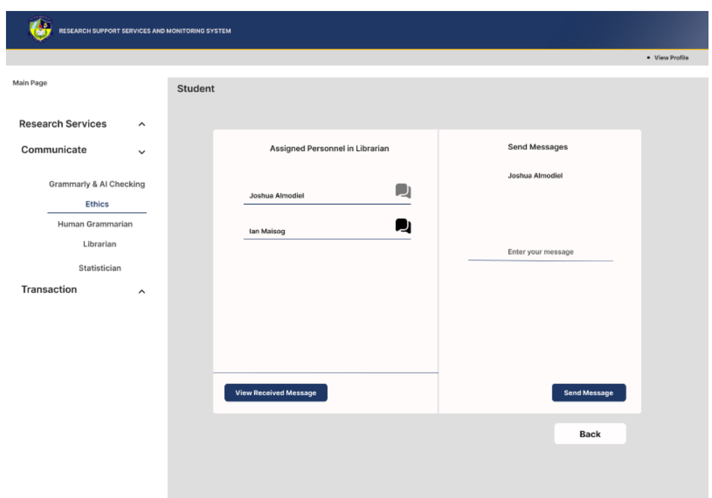

Figure 24. Students Communication to Research Services Personnel

Students can select a service to send a message to personnel assigned to that specific service student selected and view the messages they received from the personnel.

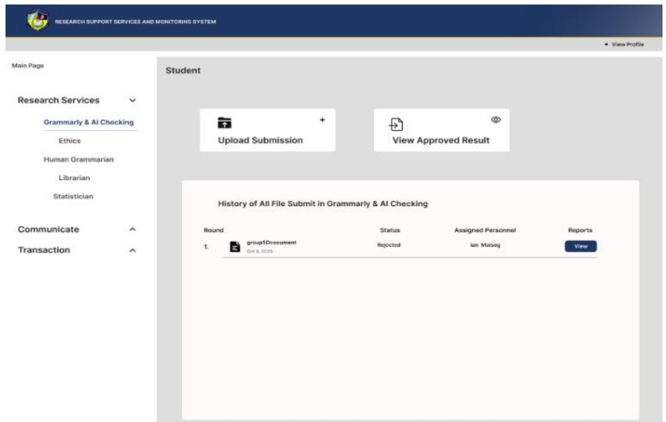

Figure 25. Student Select Grammarly & AI Checking Research Service

The student selected the Grammarly & AI Checking service, have already a record for round 1 and now submitting for the round 2. After student click the Upload Submission it will direct to transaction to pay for the round 2 submission. Only Grammarly & AI Checking research service have the transaction.

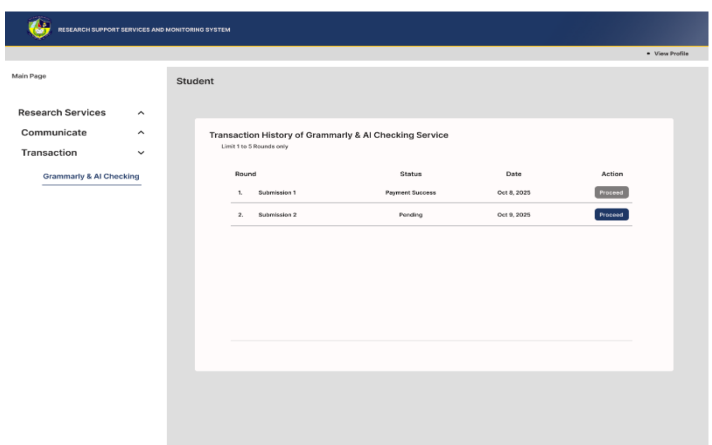

Figure 26. Student Transaction in Grammarly & AI Checking Service

This page is for student transaction history, that the student view list of the rounds submission and proceed button to direct in the payment page. Student submitting for round 2 and click the Proceed button.

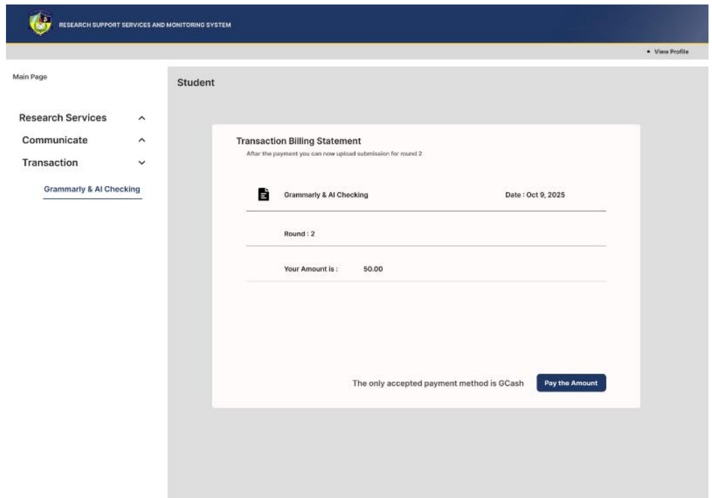

Figure 27. Student Transaction Billing Statement in Grammarly & AI Checking Service

This is the student page for the transaction billing statement, then student can view the details of the submission and pay the exact amount using Gcash payment method. After the transaction payment, student can receive an email notification and ready to upload submission for round 2.

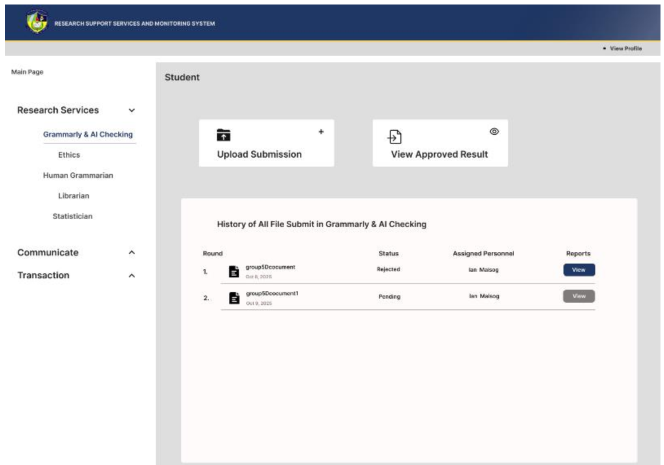

Figure 28. Student New Submission in Grammarly & AI Checking Service

Student after the transaction payment for the round 2, and it will display to the list of submission then wait for the approval. Also student can view the report for their score in Grammar & AI Checking and view comment.

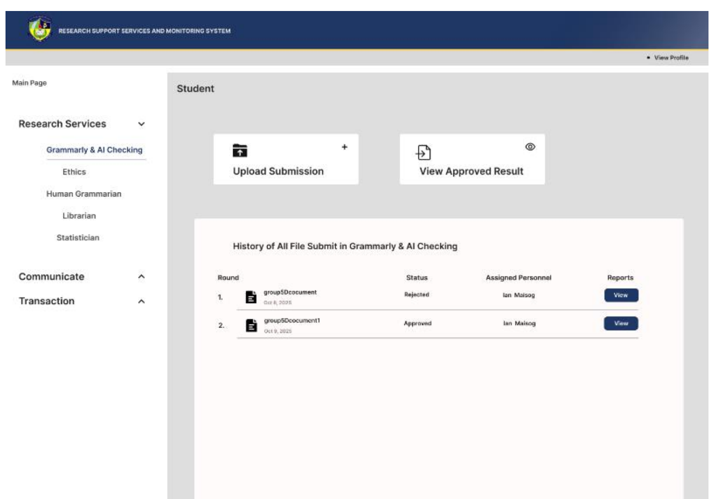

Figure 29. Student Submission Approved in Grammarly & AI Checking Service

After the round 2 submission approved, student can now view the result to view the score, certificate and can rate with feedbacks. Also the student can view reports in the list of submission. Then it will notify the student of the approval in their submission via email notification.

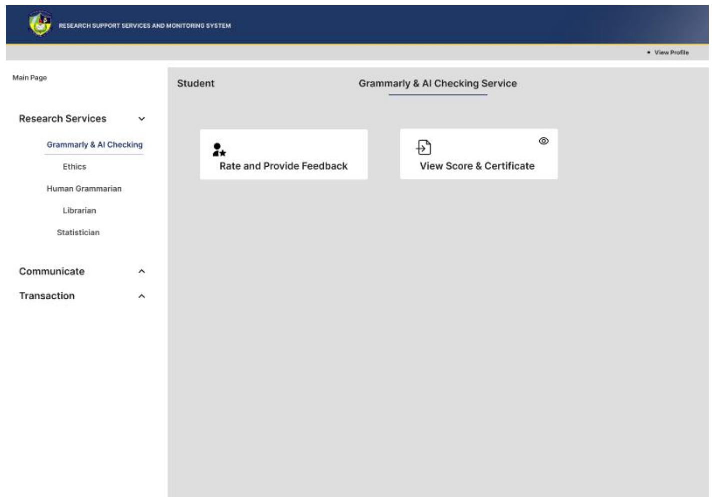

Figure 30. Student View Approved Result in Grammarly & AI Checking

Student now can provide ratings, give feedback and view score with certificate of completion after the approval of the submission.

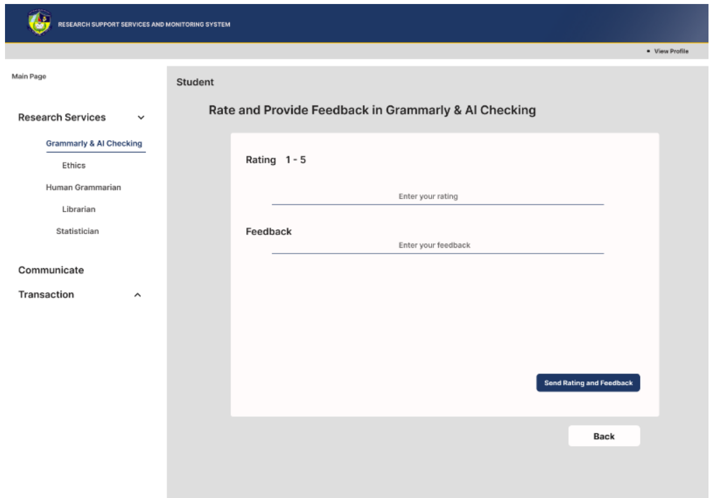

Figure 31. Students Provide Rate and Feedback

This page where the student can rate and give feedback to the personnel who handled the service.

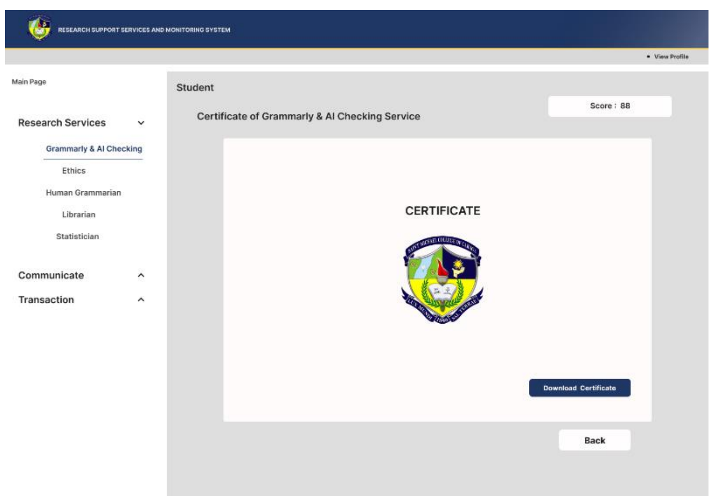

Figure 32. Students View Certificate and Score

This page allows the student to view their certificate of completion and score of the Grammarly & AI Checking service.

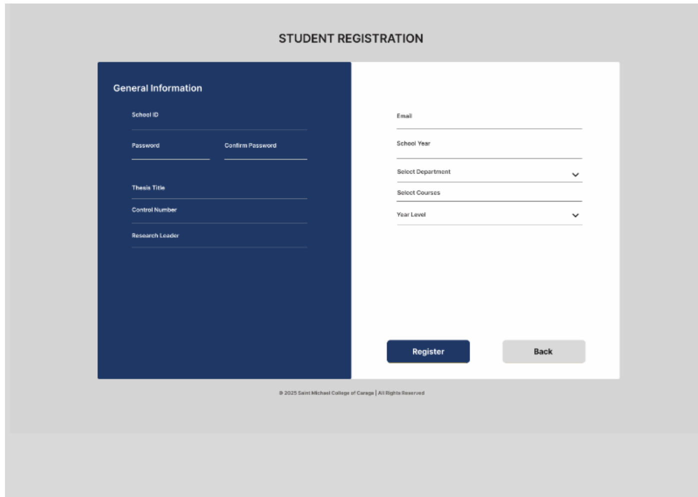

Figure 33. Student Registration

Students must enter their credentials and research information before proceeding to the login page for students' registration.

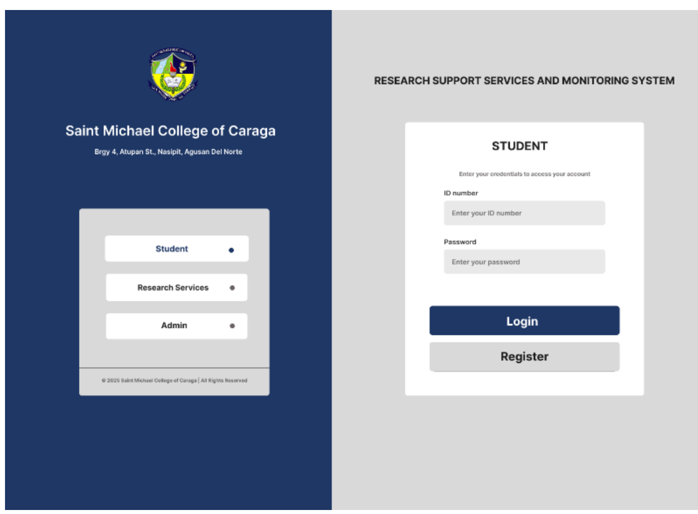

Figure 34. Grammarly & AI Checking Service Login Page

After clicking the Research Services button, users will be directed to a page where they can choose a service to log in to. Authorized personnel can then log in to access the main page of the Grammarly & AI Checking service.

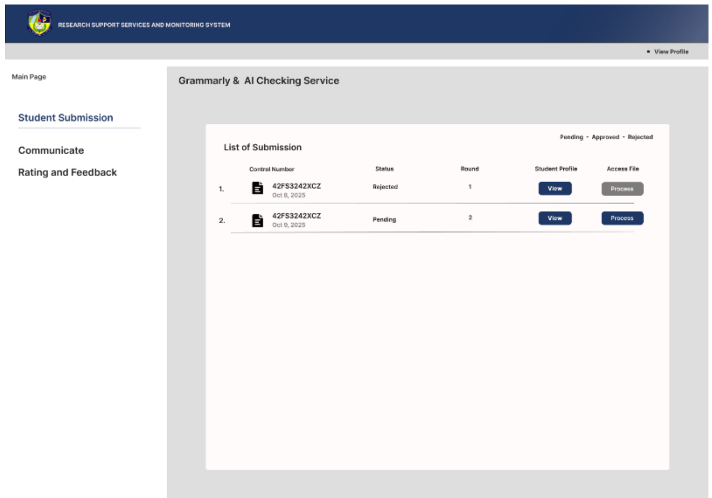

Figure 35. Grammarly & AI Checking Service Main Page

On the main page of the Grammarly & AI Checking service, personnel will see a list of research submissions from each group along with their details after logging in. From there, they can click proceed button to process the students research submissions.

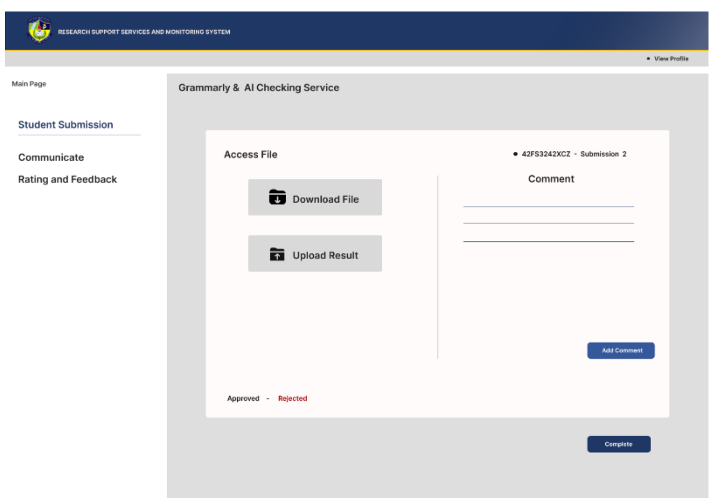

Figure 36. Grammarly & AI Checking Service Proceed to Student Submission

This is where the personnel of the Grammarly & AI Checking service will process submissions. They can access and download the files, and after processing them, upload the results or scores, add comments about the submission and change the status to approved or rejected.

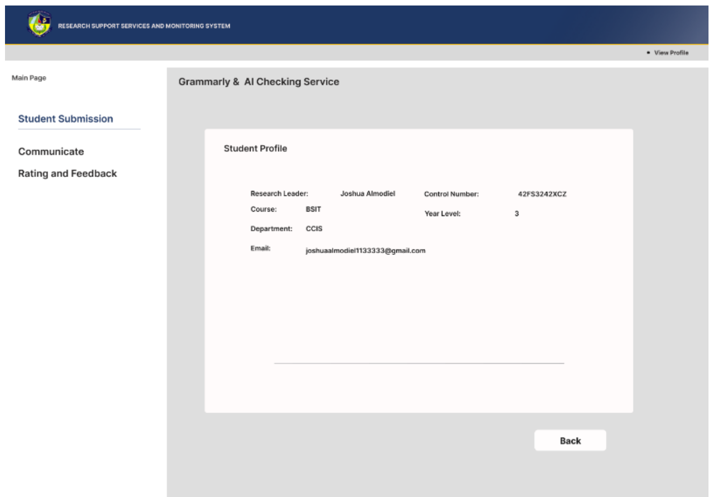

Figure 37. Grammarly & AI Checking Service Student Profile

This section displays the student profile, allowing personnel to view the student's details and research information.

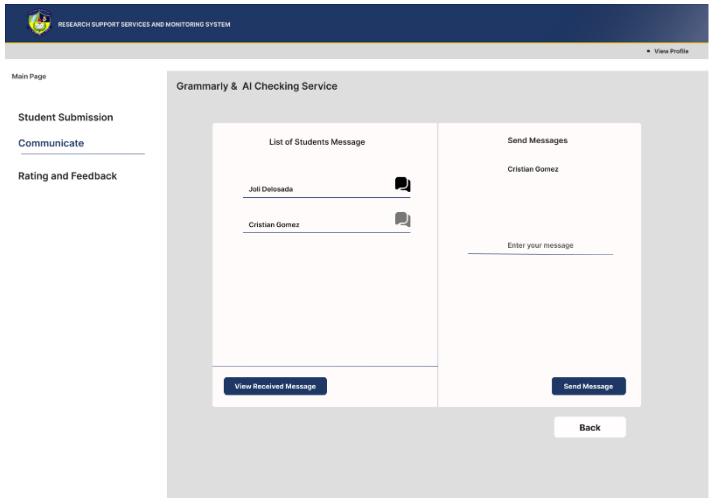

Figure 38. Grammarly & AI Checking Service Communication

In this section, personnel can view the list of messages sent by students. They can read individual messages, reply directly, or initiate new conversations to updates, or additional instructions.

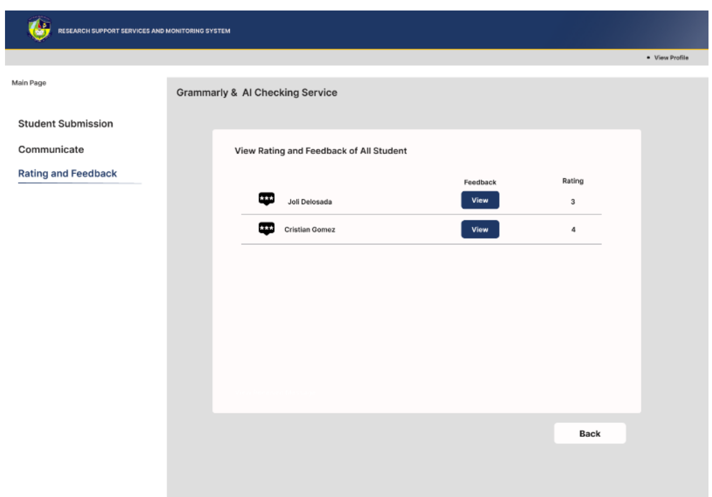

Figure 39. Grammarly & AI Checking Service Rating and Feedback from Students

In this section, personnel are able to access student ratings and review the feedback provided.

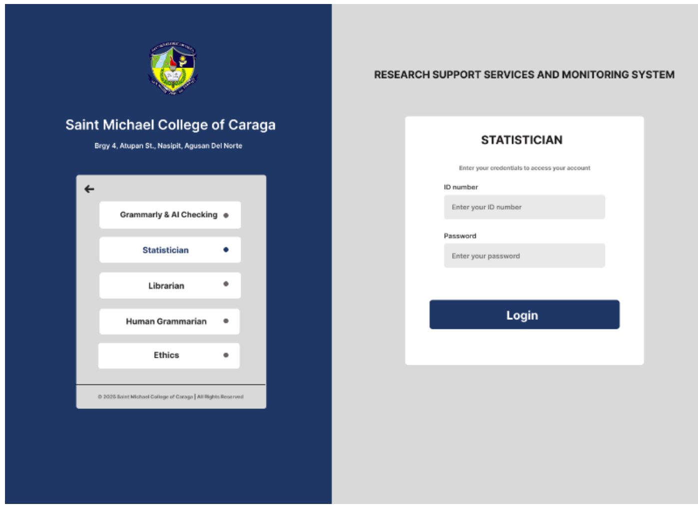

Figure 40. Statistician Service Rating Login Page

Authorized personnel can proceed to log in and access the main page of the Statistician Services.

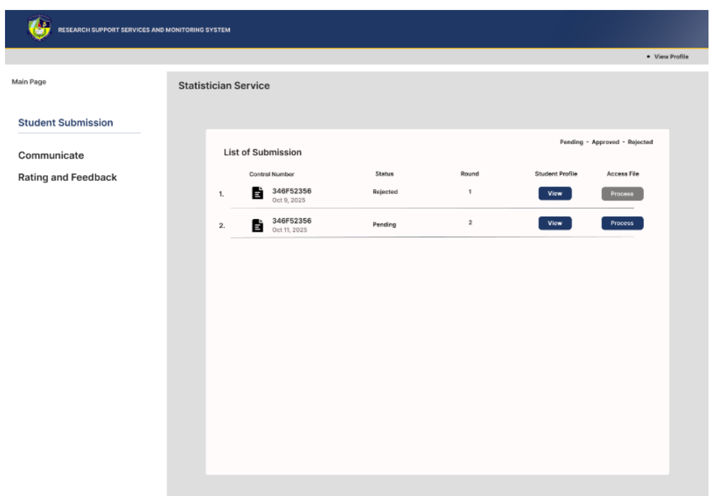

Figure 41. Statistician Service Rating Login Page

After logging in, personnel will see all group research submissions and their details on the main page. They can then process the student submissions.

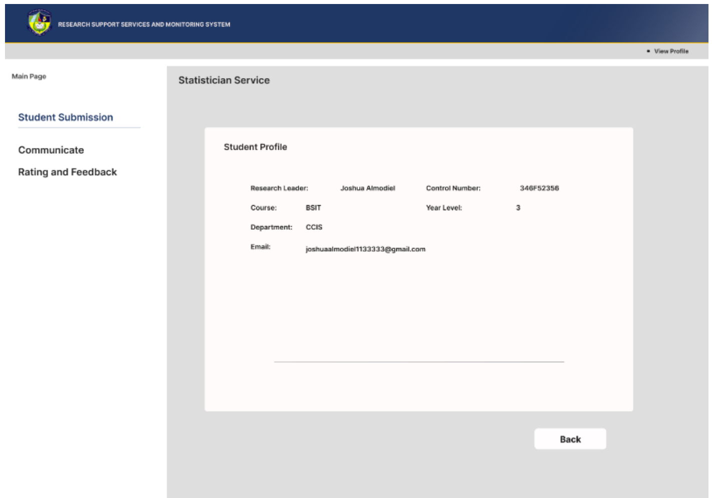

Figure 42. Statistician Service Student Profile

This section presents the student profile, enabling personnel to review the student details along with their research information.

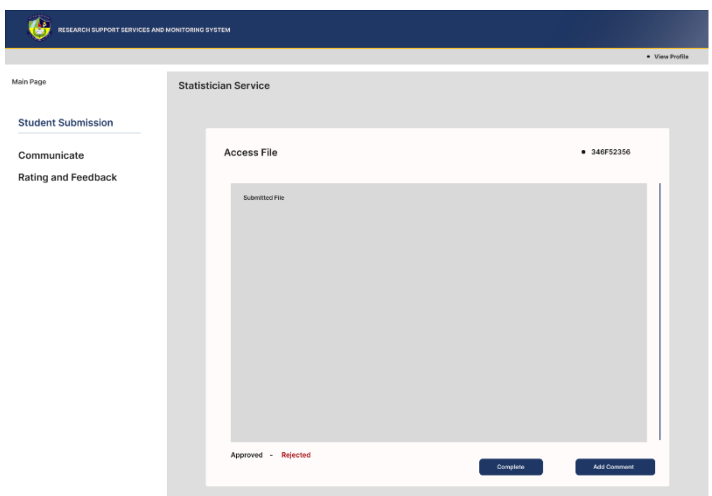

Figure 43. Statistician Service Proceed to Student Submission

This is where the personnel of the Statistician service will process submissions. They can view the file, review its contents, add comments, and once the review is complete, update the submission status to either approved or rejected.

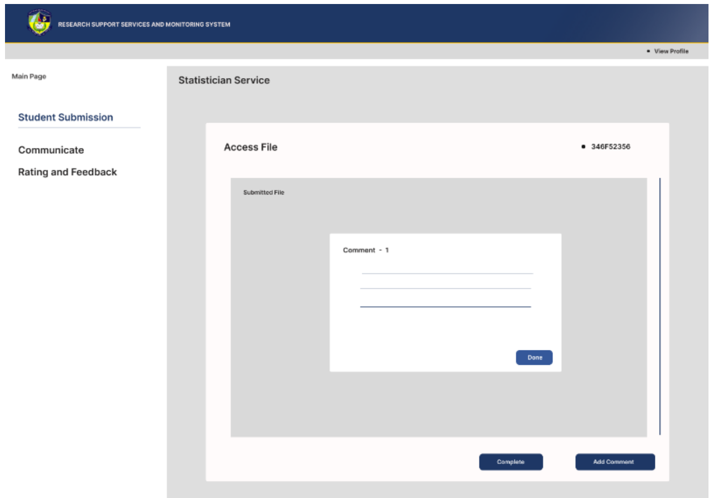

Figure 44. Statistician Service Adding Comments

In this part, personnel can add comments to the submission. They may enter Comment 1, Comment 2, and so on, depending on how many comments are necessary.

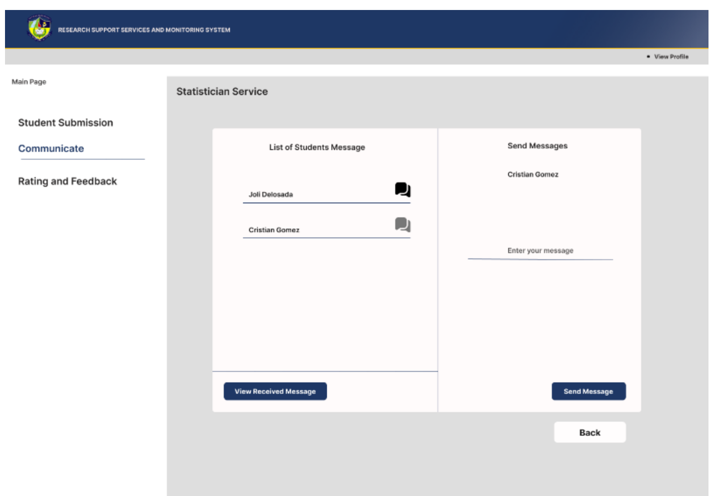

Figure 45. Statistician Service Communication

This section displays the list of messages from students, and personnel can also send replies or initiate new messages to communicate with them.

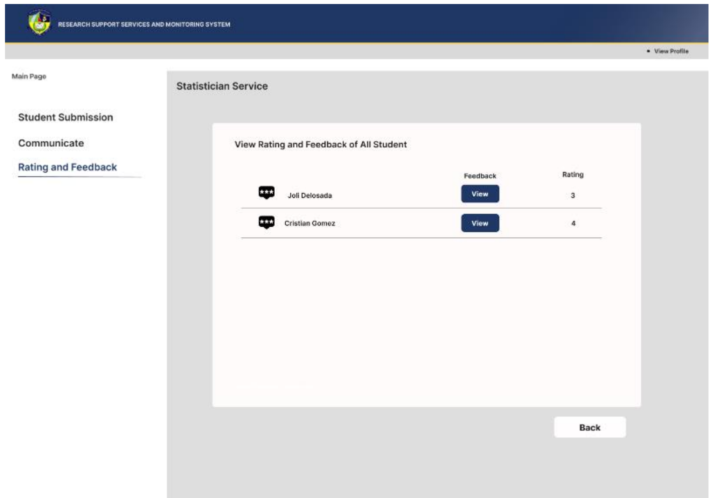

Figure 46. Statistician Service Rating and Feedback

In this section, personnel can view each student rating and read feedback regarding the performance of the personnel who handled the service.

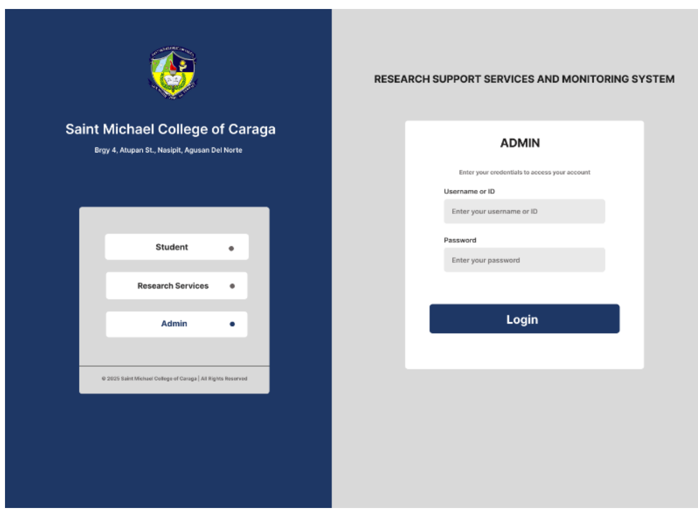

Figure 47. Admin Login Page

This is the administrator login page. After logging in, admin will be directed to the main page then manage and monitor.

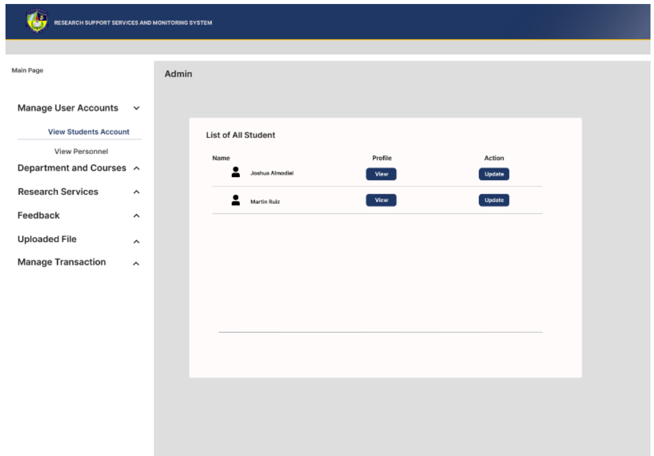

Figure 48. Admin Manage Student Accounts

This is the page where the admin can manage student accounts, also updating accounts. The admin can also view student profiles to access the details and research information.

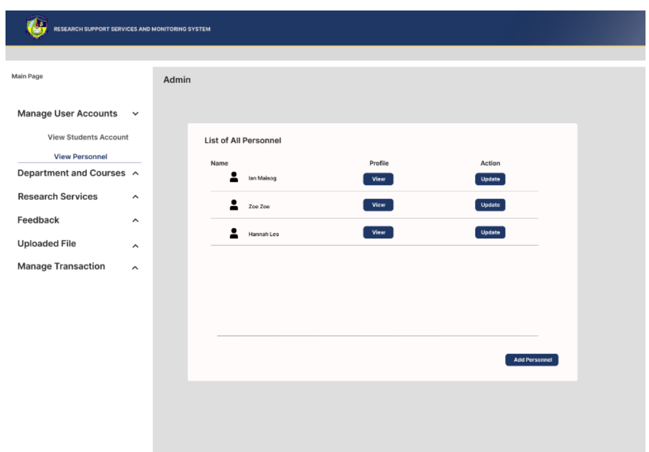

Figure 49. Admin Manage Personnel Accounts

This page allows the admin to manage personnel accounts, including updating accounts and adding new personnel. The admin can also view personnel profiles to access their research information and the services they have handled.

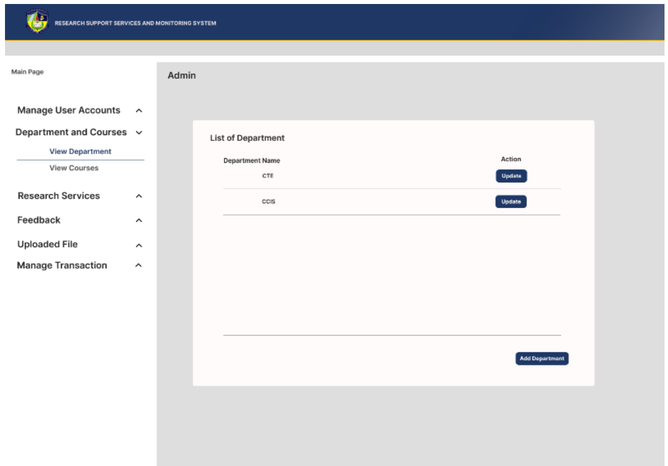

Figure 50. Admin Manage Department and Courses

This page shows the list of departments, allows admin to manage and monitor all departments and their courses. Admin can add or update departments and courses, as well as assign personnel to each department.

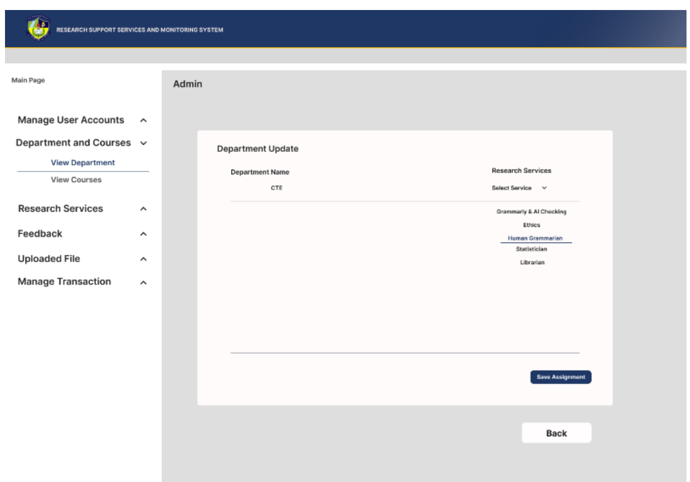

Figure 51. Admin Manage Department and Courses Selecting Service

This page allows the admin to assign personnel to a selected service within the department. The admin first selects a service and then proceeds to assign the appropriate personnel to it.

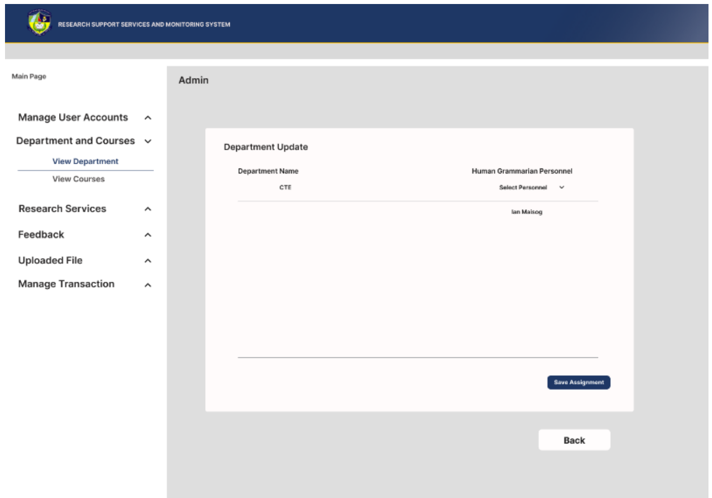

Figure 52. Admin Manage Department and Courses Assigning Personnel

This page allows the admin to select and assign personnel to a department after selecting a service.

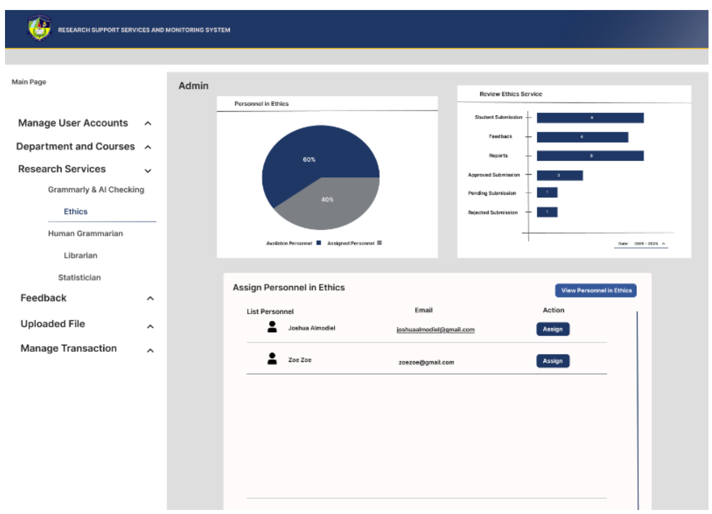

Figure 53. Admin Manage Research Service in Ethics

This page allows the admin to monitor the service and their performance, review, view the list of available personnel for assignment to the ethics service, and see the personnel currently assigned to ethics. After assigning the personnel to the service it will notify them via email.

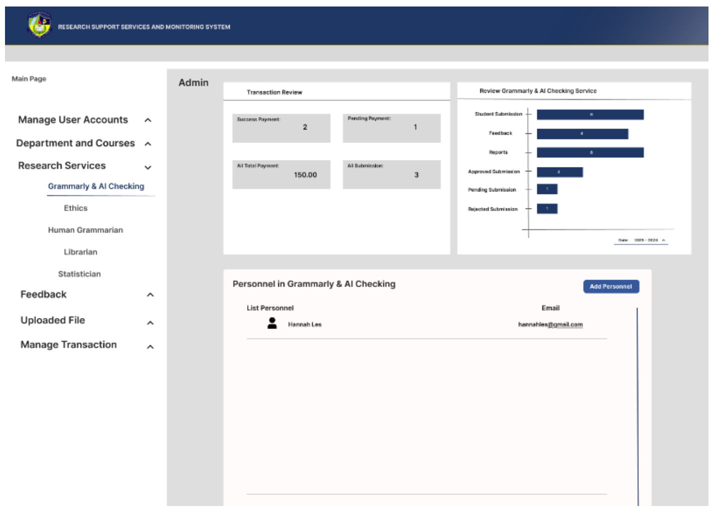

Figure 54. Admin Manage Research Service in Grammarly & AI Checking

This page is for Grammarly & AI Checking service, showing all the performance, and data review. Display the list of personnel and also can add or update. Once personnel are assign in the service, the RSSMS will notify them via email.

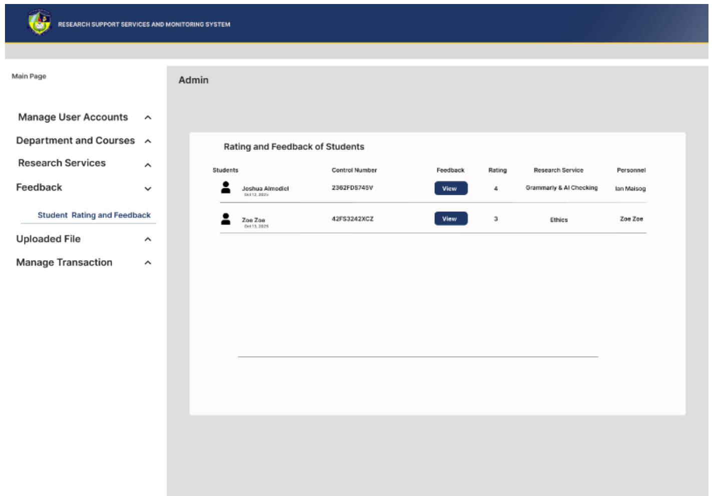

Figure 55. Admin Monitor Rating and Feedback

The admin can monitor all student ratings and feedback, as well as view additional details and research information.

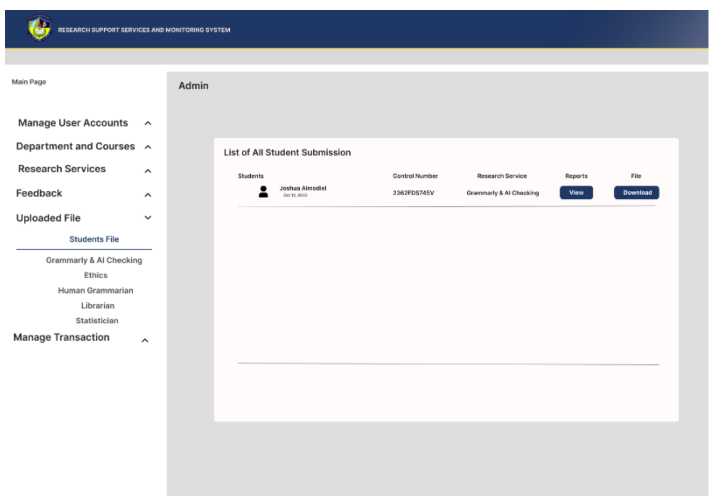

Figure 56. Admin Monitor Uploaded File of Students

This page allows the admin to monitor all submission of student, also admin can view the reports from personnel and download the file.

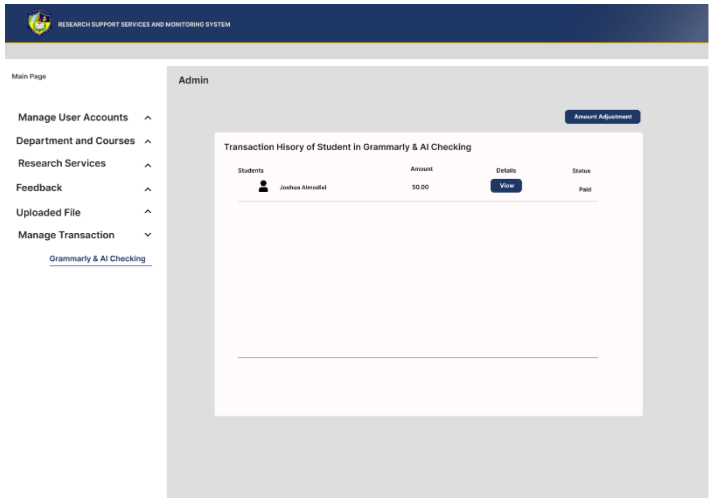

Figure 57. Admin Manage Transaction

The admin can oversee and manage Grammarly and AI Checking service transactions, with access to complete records and detailed transaction information. Also the admin can adjust the amount.

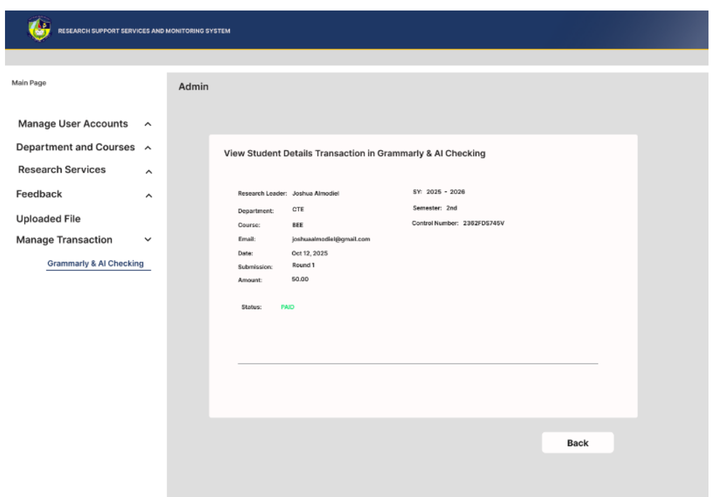

Figure 57. Admin View Student Transaction Details

This page shows detailed student transaction records, allowing the admin to review associated research information.

Table 1

Software Platforms, Development Environments, and Tools

Components	Specification	Usage
Internet Browser	Any (Chrome, Edge, Firefox, etc.)	Internet Browser serves as the primary client interface for all users of your RSSMS including Students, Admin, and Personnel. Through the browser, users access the React.js frontend, which is rendered as a dynamic, responsive web interface styled with Tailwind CSS. This browser-based design allows users to log in, upload research files, communicate with assigned personnel, view service feedback, and process GCash payments — all without installing any software.
Front-end	HTML	HTML (HyperText Markup Language) is used to design and structure the **user interface** of the RSSMS. It provides the layout for web pages such as login and registration forms, service selection pages, file submission forms, dashboards, and result viewing pages for students,

		personnel, and administrators. HTML ensures that system content is properly organized and accessible through standard web browsers.
	Tailwind CSS	Tailwind CSS is a utility-first CSS framework that simplifies the design process by providing pre-defined classes for styling. It is integrated into the RSSMS frontend to ensure a clean, modern, and responsive layout. Tailwind's flexibility allows developers to quickly style components (e.g., buttons, tables, cards, modals) while maintaining design consistency across devices. Its customization features make it easy to align the system's interface with institutional color schemes and accessibility standards, improving overall user experience.

Backend	PHP	PHP is utilized as the server-side scripting language of the RSSMS. It handles system logic such as user authentication and access control, processing of service submissions, file uploads and validation, review workflows, status updates, feedback handling, and communication between students and personnel. PHP also manages the integration with the PayMongo GCash payment gateway, verifies transactions, and controls access to result files and certificates based on payment status.
	SQL	SQL (Structured Query Language) is used to manage and manipulate data stored in the MySQL database of the RSSMS. It supports the creation, retrieval, updating, and deletion of records related to students, personnel, services, submissions, payments, feedback, and transaction history. SQL ensures data integrity through

		relational constraints and enables efficient querying for system monitoring, reporting, and administrative decision-making.
Payment Integration	GCash Payment Gateway (via PayMongo)	Integrated for secure and convenient online payment processing. It allows students to pay for their selected services directly within the RSSMS using GCash. The gateway ensures real-time transaction validation and updates payment status in the RSSMS system database.

Table 2

Hardware Requirements

Components	**Specification**	**Usage**
Desktop	Processor Intel Core i3 or higher (RAM): Minimum 4 GB Storage: 128 GB HDD or SSD Internet Connection: Reliable and stable	The device serves as the client-side hardware used by users and administrators to access the system through a web browser and perform administrative tasks, manage data, and process payments via the GCash gateway.
Server	Processor Intel Core i5 or higher (RAM): Minimum 8 GB Storage: 256 GB SSD or higher Internet Connection: Stable and fast (at least 10 Mbps)	The server is responsible for hosting the backend services, managing the database, and processing payment transactions through the GCash Payment Gateway. It ensures that all user requests and system operations run efficiently.

System Evaluation

The **Research Support Services and Monitoring System (RSSMS)** was evaluated using two approaches: the **ISO 25010 quality model**. This ensures the system is functionally reliable, user-friendly, and that its intelligent modules deliver accurate and meaningful assistance to researchers. Data was analyzed and interpreted using the following parameters:

Table 3

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional
3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Functional Suitability

Table 3 presents the criteria used to evaluate a product or System's functional suitability according to the ISO survey.

Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the System's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to assess the System's usability.

Table 6
Respondent's Distribution

SMCC Learning Resource Center		
Position	**N (Population)**	**Percentage (%)**
Personnel	5	12.5%
Student	35	87.5%
Total	40	100

Table 6 outlines the distribution of respondents who participated in evaluating the system within the SMCC Learning Resource Center. The respondents are categorized based on their positions, with 12.5% holding personnel roles and 87.5% being students. This indicates that most feedback comes from students directly interacting with the system. Since they are the primary users, their insights provide a more accurate assessment of the system's functionality, efficiency, and usability in real-world scenarios.

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD
NONE	**NONE**

ETHICAL CONSIDERATION TEMPLATE GUIDES QUESTION

A. Protection of Intellectual Property Rights (IPR)	The researchers at Saint Michael College of Caraga came up with the idea for the Integrated Research Innovation System (IRIS) to improve how they handle research. Intellectual property laws protect various outputs like interface designs, source code, documentation, and architecture, ensuring creators retain rights and promote innovation. When using external tools such as grammar checkers or plagiarism detectors, proper citation and adherence to licensing are essential. Before the implementation or system testing phase, we will explain the project's purpose, scope, and rules for handling data. The team will also clarify the ownership of all project outputs and ensure proper copyright attribution. Any third-party tools used such as React.js, Tailwind CSS, Node.js, MongoDB, and the GCash API will be acknowledged according to their respective open-source licenses. The project will comply with intellectual property rights by giving credit to original creators and specifying which works belong to the institution or to the

developers. Following best practices maintain legal compliance, transparency, and integrity, helping organizations avoid disputes and support a fair, innovative environment. The project honors the idea that an institution owns the system and ensures people give credit to anyone whose work or ideas they use.

B. Informed Consent

Before the implementation or system testing phase, We will explain the project's purpose, scope, and rules for handling data to all participants before starting any system testing. The team will clearly explain the project's purpose, procedures, and how participant data will be used and protected. Participants will be informed about their rights and the voluntary nature of their involvement. Written informed consent will be obtained from all participants before gathering any information or feedback. They will also be informed that they may refuse or withdraw from the activity at any time without penalties or consequences. This approach ensures that developers and institutional partners maintain clarity and effective communication.

C. Data Privacy and Confidentiality

The IRIS will follow the Data Privacy Act of 2012 (RA 10173) to keep all information private and secure. Only authorized personnel specifically the Research Director and the System Administrator (superuser) may access and modify sensitive research data stored in the system. To protect these records, the system will use encryption, database security, user authentication, access controls, and data anonymization. These measures ensure that only allowed users can access the data and that identities remain protected. A retention policy will also guide how long information is stored and managed.

D.Voluntary Participation and Freedom to Withdraw

Participation of faculty and students in system testing will be voluntary, with clear informed consent provided. They may withdraw at any time without any consequences. The team will ensure no coercion, and participants will be informed of their rights and freedom throughout the process.

E. Minimization of Harm and Risk Management

The team will use the IRIS system carefully to ensure user safety and prevent potential risks such as data loss, security issues, or system errors. Clear guidelines will be followed to protect user information and uphold ethical standards. The researchers will thoroughly test all features to avoid misuse and ensure accuracy. Technical support and training will be provided to help users address concerns during the pilot phase and reduce mistakes.

F. Beneficence and Contribution to Knowledge

This project ensures beneficence by providing clear benefits to both participants and the institution. Students, advisers, and research staff will benefit from automated tools such as plagiarism checking, grammar review, and analytics dashboards, which make research tasks faster, more accurate, and easier to manage. These features also give timely feedback and improve overall research productivity. For Saint Michael College of Caraga, the system strengthens institutional research processes by centralizing workflows, promoting transparency, and supporting adviser student collaboration. The results of the system will be shared with the SMCC Research Office and may serve as a basis for future improvements. These contributions show that the project supports both participant welfare and institutional academic development. Model for future institutional innovations.

G. Justice and Fair Participant Selection

Participants will include faculty advisers, students, and research staff who are directly involved in the research activities. The selection will follow clear inclusion and exclusion criteria to ensure fairness. Only individuals who are academically involved in the IRIS project, available during data collection, and capable of providing relevant feedback will be included. No participant will be excluded based on personal characteristics such as age, gender, socioeconomic status, or any social factor. The selection process ensures equal opportunity for all eligible stakeholders to contribute, aligning with the principles of fairness, transparency, and justice. This approach guarantees that participation is based solely on relevance to the project and not on preference or bias. Ensuring that all stakeholders have equal opportunity to participate, provide input, and benefit from the development and implementation of the system fosters inclusivity and collaboration.

H. Data Integrity and Accuracy

The system will enforce data validation, monitoring, and version control to maintain information accuracy. To ensure consistency, the research team will verify progress tracking and documentation The researchers will honestly present all reports, reviews, and test results from the IRIS project. They will disclose any problems, errors, bugs, system limitations, or constraints encountered during development. Full and accurate reporting will be ensured to maintain transparency and integrity throughout the study. Additionally, any AI tools used during the writing, checking, or refinement of the document will be properly acknowledged in accordance with institutional guidelines. adviser dashboards and automated logs. Regular audits and feedback sessions will help maintain reliable, updated, and tamper-free data. The team will promptly correct any identified errors to uphold the integrity of institutional research records.

I. Transparency and Honesty in Reporting

The researchers will honestly present all reports, reviews, and test results from the IRIS project. They will document any problems, restrictions, or bugs in the system that arise during development appropriately. The researchers will honestly present all reports, reviews, and test results from the IRIS project. They will disclose any problems, errors, bugs, system limitations, or constraints encountered during development. Full and accurate reporting will be ensured to maintain transparency and integrity throughout the study. Additionally, any AI tools used during the writing, checking, or refinement of the document will be properly acknowledged in accordance with institutional guidelines.

J. Use of Patented or Copyrighted Materials

Third-party tools or libraries integrated into the system such as React.js, Tailwind CSS, Node.js, MongoDB, and the GCash API will be used under their respective open-source or proprietary licenses. The researchers will ensure full compliance with the licensing terms, permissions, and intellectual property policies associated with these technologies. Proper attributions will also be included in the documentation to avoid copyright infringement. The team will strictly follow legal and ethical standards in using, modifying, and integrating these resources throughout the development process. The researchers will include proper citations and attributions in the documentation to avoid copyright infringement. They ensure compliance with intellectual property laws and demonstrate academic and ethical responsibility in the use and development of software.

K. Considerations for Human Subjects

This study involves human participants, including faculty advisers, students, and research personnel who will test or provide feedback on the system. No physical or psychological harm will occur during participation. The researchers will obtain informed consent from all human participants to ensure that their rights, privacy, and confidentiality are fully protected. Data collection will focus solely on research-related processes, ensuring that all personal and academic information remains confidential throughout all phases of the study. The research team will strictly adhere to ethical standards regarding respect, consent, and privacy. No animal subjects are involved in this study. The project does not require any form of animal testing, and therefore no risks or procedures involving animals are present.

M. Responsible Use of AI and Other Technologies

Responsible Use of AI Tools AI tools such as ChatGPT and Grammarly may assist in improving the clarity, organization, and grammar of the written document. However, these tools did not alter, manipulate, or generate any research data. All data collection, analysis, and validation were carried out solely by the researchers. AI tools were used only as support for writing and idea refinement, not for verifying or modifying research findings. This ensures responsible AI use and maintains the accuracy, reliability, and integrity of the study.

Ethical Clearance and Institutional Approval The project will secure ethical approval from the Institutional Review Board (IRB) / Research Ethics Committee of Saint Michael College of Caraga. This approval is required to ensure that the study is ethically sound before any data collection or system implementation begins. The clearance will confirm that the research protects participant privacy, ensures informed consent, and follows proper data-security procedures. By obtaining ethics approval, the researchers demonstrate accountability and compliance with institutional, national, and international ethical standards throughout the entire duration of the study.